

Bell-Decomposition Quantum Kernels: A Multi-Scale Benchmark on Earth Observation and Beyond

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Abstract

We introduce the Bell-decomposition family of two-qubit feature maps, unifying the singlet-only Self-Regulating Quantum Feature Map (SRQFM) and the four-Bell-sector Bell-Spectrum Coupling Map (BSCM) under an adjustable prior. As a specialisation of the Pauli-decomposition formalism of Incudini et al. to the Bell subspace, the family admits two closed-form structural properties: a tight bound $|\alpha| \leq 1/2$ on every per-pair Pauli coefficient, and $\alpha_{ZZ} \equiv 0$ for the uniform prior, making BSCM-uniform strictly Pauli-orthogonal to any SRQFM-style ZZ generator. A single hyperparameter τ , locked once at $n = 8$, transfers operationally to $n = 14$ and $n = 16$. We report four findings. (i) **Architectural advantage**: at 16 qubits and $N = 10,000$ the Bell-decomposition projected kernels add $+0.167$ – $+0.170$ macro-F1 over commuting ZZ -PQK on So2Sat LCZ42 (17 classes) and $+0.097$ – $+0.108$ on EuroSAT (10 classes); Holm-corrected Wilcoxon $p = 0.0078$ on both. (ii) **Domain-agnostic**: the gap is positive on every split across four UCI datasets (largest $+0.646$ accuracy on the 26-class Letter task, whose 16 features match $n = 16$) and on a transverse-field Ising-model phase-classification benchmark. (iii) **Noise-robust**: under density-matrix simulation at $n = 8$ the gap remains $+0.133$ – $+0.136$ across $p \in \{0, 0.001, 0.005, 0.01\}$. (iv) **Concentration on real data**: a 14-qubit BSCM build at $N = 800$ and a 16-qubit depth scan at $N = 2,000$ reproduce the Thanasilp et al. concentration signature on real multi-class remote-sensing data. The Bell-decomposition kernels match but do not exceed tuned classical RBF-SVM at any scale tested, consistent with recent theory (Heyraud et al.; Slattery et al.) and prior empirical work (Bowles et al.; Schnabel and Roth).

Keywords: Quantum kernel methods, Pauli decomposition, Bell-state feature maps, Projected quantum kernels, Exponential concentration, Land cover classification

1 Introduction

Quantum machine learning (QML) has been studied as a possible route to computational advantage in pattern recognition. Among the proposed approaches, kernel methods are particularly clean: a parameterised unitary $U(x)$ embeds a classical input into a quantum feature space, the fidelity kernel $K(x, x') = |\langle 0|U^\dagger(x')U(x)|0\rangle|^2$ defines a similarity function, and a downstream classical Support Vector Machine (SVM) does the learning [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#). Theoretical analysis has shown that for problems whose hardness matches the circuit’s inductive bias, quantum kernels can in principle outperform any efficient classical kernel [Liu et al. \(2021\)](#). However, for typical applied datasets this match is rare, and recent benchmarks — Bowles *et al.* [Bowles et al. \(2024\)](#) on 160 binary tasks, Schnabel & Roth [Schnabel and Roth \(2025\)](#) on 64 datasets in QMI 2025 — consistently fail to find a robust advantage over tuned classical baselines such as the radial-basis-function (RBF) kernel SVM.

Land cover classification from satellite imagery is a stringent applied benchmark: So2Sat LCZ42 [Zhu et al. \(2020\)](#) contains 400,673 multimodal Sentinel-1 (synthetic-aperture-radar, SAR) and Sentinel-2 (multispectral optical) patches across 42 cities and 17 Local Climate Zone (LCZ) classes, with class fractions ranging from 0.7% to 13.6% in our subsample; EuroSAT [Helber et al. \(2019\)](#) contains $\sim 27,000$ Sentinel-2 patches in 10 classes. Despite growing QML interest in Earth observation [Rodriguez-Grasa et al. \(2025\)](#), no published kernel-method study we are aware of has evaluated multiple quantum kernel architectures on these benchmarks at 16 qubits with $N = 10,000$ under a fixed-pool, multi-seed, C -tuned protocol — existing QK benchmarks (Schnabel & Roth [Schnabel and Roth \(2025\)](#), $N \leq 500$ on 64 mostly-synthetic datasets; Bowles *et al.* [Bowles et al. \(2024\)](#), 160 binary tasks; Rodriguez-Grasa *et al.* [Rodriguez-Grasa et al. \(2025\)](#), $N \leq 1,800$ binary solar-panel detection) operate at smaller scale or different class structure. This paper supplies the evaluation at the scale our contribution requires.

The technical content is organised around a single design principle. The two-qubit encoded state $|\psi(x_i)\rangle|\psi(x_j)\rangle$ can be written in the Bell basis as a weighted sum of four projectors, $|\Phi^\pm\rangle\langle\Phi^\pm|$ and $|\Psi^\pm\rangle\langle\Psi^\pm|$, with weights determined by the inputs. Choosing which weights are activated as data-dependent gate angles maps out a structured design space. Two natural choices populate this space: SRQFM (singlet-only, equivalent to a particular ZZ -fidelity construction) and BSCM (all four sectors with an adjustable prior). The two constructions are theoretically distinct — the *uniform* BSCM preset is provably Pauli-orthogonal to SRQFM in the per-pair generator (§3.3) — but turn out to be empirically indistinguishable in macro-F1 at $N \leq 2000$. The interesting behaviour appears at the boundaries: at 14 qubits both members regress toward concentrated kernels with weakened class structure; at 16 qubits and $N = 10,000$ the Bell-decomposition *projected* quantum kernels (BSCM-PQK family) add +0.167 to +0.170 macro-F1 over the canonical Standard ZZ -PQK on So2Sat

and +0.097 to +0.108 on EuroSAT, demonstrating that non-commuting gate geometries substantially outperform the canonical commuting ZZ architecture within the projected quantum kernel framework. On four standard UCI datasets (Digits, Breast Cancer, Letter Recognition with 16 features exactly matching $n = 16$, and Wine as a 13-feature padding boundary case) BSCM-uniform-PQK exceeds Standard- ZZ -PQK on every dataset (gap +0.546 on Digits, +0.289 on Breast Cancer, +0.646 on Letter, +0.348 on Wine), confirming that the architectural advantage is domain-agnostic; BSCM-PQK additionally *matches* classical baselines within ~ 2 percentage points on Digits and Breast Cancer. All headline architectural-advantage numbers in this paper refer to the projected-kernel readout (Bloch-vector RBF) unless explicitly labelled *fidelity* or *FQK*; fidelity-kernel results for the same family appear in the smaller-scale $N \leq 2,000$ experiments E26/E32.

We make four claims:

1. **Family.** The Bell-decomposition family unifies SRQFM and BSCM as Pauli-decomposition complements with a closed-form per-pair generator and provably bounded gate angles ($|\alpha| \leq 1/2$ for the three priors evaluated in this paper, uniform / phi-only / psi-only; Theorem 1(a)). A single hyperparameter τ controls the coefficient scale; the value $\tau = 0.25$ locked from a 200-sample holdout at $n = 8$ transfers operationally to $n = 14$ and $n = 16$ without re-tuning (§3.3).
2. **Architectural advantage.** At 16 qubits and $N = 10,000$ on So2Sat LCZ42 and EuroSAT, the Bell-decomposition *projected* kernels (BSCM-PQK family) add +0.167 to +0.170 macro-F1 over the canonical commuting Standard ZZ -PQK on So2Sat and +0.097 to +0.108 on EuroSAT, with Holm-corrected Wilcoxon $p = 0.0078$ on both datasets (the 10-split minimum achievable raw $p = 1/2^9 \approx 0.002$). Non-commuting XX , YY , ZZ gate geometries materially outperform the single-axis ZZ within the projected quantum kernel framework.
3. **Domain-agnostic.** On four standard UCI tabular datasets — Digits, Breast Cancer, Letter Recognition (16 features exactly matching $n = 16$), and Wine (13 features, padded as a boundary case) — and one intrinsically-quantum dataset (TFIM phase classification on 16-site ground states), BSCM-uniform-PQK exceeds Standard- ZZ -PQK by a positive margin on every dataset and every split. Standard- ZZ -PQK collapses on every evaluated UCI dataset (41.4%, 61.6%, 13.3%, 40.7% accuracy on Digits, Breast Cancer, Letter, Wine, versus BSCM’s 96.0%, 90.5%, 77.9%, 75.6%), with the largest architectural gap +0.646 accuracy on the 26-class Letter task. BSCM-vs-classical: ≤ 2 percentage points on Digits and Breast Cancer; ~ 8 pts behind RBF on the 26-class Letter task; ~ 21 pts on Wine, attributable to the $n = 16$ -vs-13-feature circuit-width artefact analysed in §5.3. The Earth-observation-specific-artefact reading is therefore ruled out.
4. **Concentration on real data.** Two complementary measurements (real-data 14-qubit BSCM at $N = 800$ on physics-Fisher-14; 16-qubit depth scan over $L \in \{2, \dots, 10\}$ at $N = 2,000$) exhibit the canonical Thanasilp concentration signature Thanasilp et al. (2024) on real multi-class remote-sensing data, with all four kernel-health diagnostics moving in the predicted directions and a corresponding regression of macro-F1 toward tuned classical baselines.

We make no claim of quantum advantage: at every scale tested, no Bell-decomposition kernel beats both tuned RBF-SVM and Random Forest; tuned Random-Fourier-Feature dequantisation [Rahimi and Recht \(2007\)](#) of the standard ZZ-FQK in fact *exceeds* it by +0.010 macro-F1 on So2Sat PCA-8 at $D = 512$ features (§5.7). This match-not-beat outcome is the expected regime under [Heyraud et al. \(2022\)](#); [Slattery et al. \(2025\)](#) and consistent with [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#); [Rodriguez-Grasa et al. \(2025\)](#) at smaller scale.

What is new. The general Pauli-decomposition formalism for fidelity kernels is due to Incudini *et al.* [Incudini et al. \(2025\)](#); our Theorem 1 is a specialisation of that formalism to the Bell-projector subspace, and yields two closed-form structural identities not available from the general formalism: **(a)** an explicit, tight bound $|\alpha_{XX}|, |\alpha_{YY}|, |\alpha_{ZZ}| \leq 1/2$ valid for all three priors and saturated on the input cube boundary; **(b)** the closed-form identity $\alpha_{ZZ} \equiv 0$ for the uniform prior, establishing strict Pauli-orthogonality between BSCM-uniform and any SRQFM-style ZZ-generator. In addition we report **(c)** an *empirical regularity* not predicted by the general formalism — a single scalar hyperparameter τ , locked once from a 200-sample holdout at $n = 8$, transfers operationally to $n = 14$ and $n = 16$ without re-tuning and produces both the predicted concentration regression (B2 / E37) and the headline architectural-gap result (E38) at the larger qubit counts (§3.3; we do not claim $\tau = 0.25$ is the argmax of KTA at $n = 14$ or $n = 16$, only that it remains operationally well-conditioned). Property (c) is presented as an empirical observation throughout, not as part of Theorem 1; it is consistent with — and partially motivated by — the bound in (a), but the scale transferability across qubit counts is verified experimentally, not proven. Beyond the analysis, no prior study we are aware of has (d) measured the Thanasilp [Thanasilp et al. \(2024\)](#) concentration regression on real multi-class remote-sensing data at 14 qubits with four simultaneously monitored kernel-health diagnostics, or (e) benchmarked multiple quantum kernel architectures at 16 qubits and $N = 10,000$ on Earth-observation data under a fixed-pool, multi-seed, C -tuned protocol with τ -locked hyperparameters and Holm-corrected significance tests.

Relation to recent benchmarks. Our finding that bandwidth-tuned PQKs match (but do not exceed) tuned classical RBF-SVM is the expected outcome of recent theory: Heyraud *et al.* [Heyraud et al. \(2022\)](#) showed that PQKs converge to RBF behaviour at optimal bandwidth, and Slattery *et al.* [Slattery et al. \(2025\)](#) formalised this similarity for arbitrary feature maps. Empirical benchmarks have reached the same conclusion at smaller scale ([Schnabel and Roth \(2025\)](#), 64 datasets, $N \leq 500$; [Bowles et al. \(2024\)](#), 160 binary tasks; [Rodriguez-Grasa et al. \(2025\)](#), neural quantum kernels $\approx$ Random Forest at 88% on a binary EO task). Our contribution is therefore not a new finding of *whether* quantum kernels match classical, but: (i) the Bell-decomposition family of fidelity feature maps and the quantum-vs-quantum advantage of non-commuting gate geometries over the canonical ZZ architecture, (ii) directly measured concentration regression on real multi-class data at 14 qubits, and (iii) domain-generalisation evidence showing that the architectural advantage is not specific to Earth observation, and (iv) a fixed-protocol multi-class Earth-observation benchmark at 16 qubits and $N = 10,000$ that, to our knowledge, is the largest of its kind.

2 Related Work

Quantum kernel theory.

Havlíček *et al.* [Havlíček et al. \(2019\)](#) introduced the ZZ Feature Map. Liu *et al.* [Liu et al. \(2021\)](#) proved an exponential advantage for a discrete-log task under specific hardness assumptions. Huang *et al.* [Huang et al. \(2021\)](#) proposed the geometric difference $g(K_q, K_c)$ as a separation diagnostic and showed advantage requires both $g \gg 1$ and positive ΔKTA . Schuld [Schuld \(2021\)](#) showed angle-encoded kernels are Fourier-series approximations whose expressibility beyond classical kernels requires entanglement.

Concentration and projected kernels.

Thanasilp *et al.* [Thanasilp et al. \(2024\)](#) proved that fidelity kernels with global observables undergo *exponential concentration*: off-diagonal entries converge exponentially in qubit count to a constant, eliminating discriminative structure. The Projected Quantum Kernel (PQK) [Huang et al. \(2021\)](#) extracts single-qubit Bloch expectations and applies an RBF on the resulting low-dimensional embedding; Shaydulin & Wild [Shaydulin and Wild \(2022\)](#) suggested CV-KTA bandwidth selection, which we adopt. Heyraud *et al.* [Heyraud et al. \(2022\)](#) analysed PQKs under depolarising noise and showed that the “classical-like” regime of PQKs persists under realistic gate-error rates — a result we use in our noise-robustness analysis (§5.7). Slattery *et al.* [Slattery et al. \(2025\)](#) further formalised the conditions under which bandwidth-tuned quantum kernels become statistically indistinguishable from classical kernels.

Benchmarking.

Bowles *et al.* [Bowles et al. \(2024\)](#) found classical models match or exceed twelve QML models on 160 binary tasks. Schnabel & Roth [Schnabel and Roth \(2025\)](#) reached the same conclusion across 64 datasets, 9 encodings, and $\sim 20,000$ models in QMI 2025. Rodriguez-Grasa *et al.* [Rodriguez-Grasa et al. \(2025\)](#) reported *competitive but not superior* neural quantum kernel performance on binary solar-panel detection in MLST 2025. Table 1 places the present study in context. None of the cited benchmarks combines multi-class classification ($K \geq 10$), $n = 16$ qubits, and $N = 10,000$ samples in a fixed-protocol study on Earth-observation data; our work fills that gap.

Table 1 Scale comparison of recent fixed-protocol quantum-kernel benchmarks. n =qubits (maximum reported), N =pool size, EO=Earth observation. Our study (bottom row) is the only one combining $n \geq 16$, $N \geq 10,000$, and $K \geq 10$ on EO.

Study	n	N	K	Domain
Schnabel & Roth Schnabel and Roth (2025)	≤ 8	≤ 500	2	Mixed tabular
Bowles <i>et al.</i> Bowles et al. (2024)	≤ 16	≤ 500	2	Mixed
Rodriguez-Grasa <i>et al.</i> Rodriguez-Grasa et al. (2025)	≤ 8	$\leq 1,800$	2	EO (solar)
This work	16	10,000	10–17	EO (So2Sat, EuroSAT)

Bell- and Pauli-decomposition feature maps.

Suzuki *et al.* Suzuki et al. (2020) introduced Pauli-rotation feature maps with arbitrary two-body generators. Incudini *et al.* Incudini et al. (2025) formalised fidelity kernels as arbitrary-weight Pauli-basis expansions of the form $H_{ij} = \sum_{P \in \{X,Y,Z\}^{\otimes 2}} \alpha_P P_i P_j$ and derived a closed-form expression for the resulting kernel matrix in terms of the α_P coefficients. That work is generator-agnostic by design: given any data-dependent choice of the α_P , it tells you what the kernel matrix will look like.

Bell-projector specialisation.

The general Incudini formalism gives a closed-form kernel matrix as a function of arbitrary α_P but does not constrain those coefficients — so for unconstrained H_{ij} the gate angles can diverge, the formalism is silent on which priors produce Pauli-orthogonal families, and scale-stability across n is unaddressed. Restricting H_{ij} to a non-negative-prior weighted sum of Bell projectors yields three properties not derivable from the general formalism: (a) an explicit prior-independent tight bound $|\alpha_P| \leq 1/2$, saturated on the boundary of $[0, \pi]^2$ (Theorem 1(a)), following from $w_k \in [0, 1/2]$ and the $\{\pm 1/4\}$ projector coefficients of Lemma 1; (b) the closed-form identity $\alpha_{ZZ} \equiv 0$ for the uniform prior $p = (1, 1, 1, 1)$ (Theorem 1(b)), from the sector-sum identities $w_{\Phi+} + w_{\Phi-} = w_{\Psi+} + w_{\Psi-} = 1/2$; and (c) an empirically scale-stable single hyperparameter τ , locked at $n = 8$ and operational at $n = 14$ and $n = 16$ without re-tuning (Tabs. 9, 10, 11). Properties (a) and (b) are proven; (c) is an empirical regularity consistent with (a), not a theorem. All three are subspace-specific; in the framing of Incudini et al. (2025) the contribution is new *syntax* (a canonical four-point design space with closed-form bounds, an orthogonality identity, and a scale-stable hyperparameter), unit-tested against the general formalism numerically (`tests/test_proofs.py`, 10^{-8} tolerance). SRQFM and BSCM are two specific points in this design space.

Inductive bias and bandwidth.

Kübler *et al.* Kübler et al. (2021) showed angle-encoded quantum kernels possess a high-frequency inductive bias detrimental on smooth targets — precisely the regime occupied by spectral remote-sensing features. Canatar *et al.* Canatar et al. (2023) proved bandwidth selection is critical for generalisation.

Remote sensing benchmarks.

On So2Sat LCZ42 Zhu et al. (2020), deep CNN baselines (Sen2LCZ-Net Qiu et al. (2020), ResNeXt) reach ~ 61 – 70% overall accuracy on the held-out cultural-10 test split using the full 32×32 image cube. EuroSAT Helber et al. (2019) is essentially saturated by deep models ($\geq 98\%$ with ResNet-50). Tabular handcrafted-feature pipelines score lower; their interest is methodological — they provide a controlled, low-dimensional setting in which kernel-vs-kernel separation can be measured cleanly.

3 Methods

3.1 Data and feature engineering

We use two datasets. **So2Sat LCZ42** [Zhu et al. \(2020\)](#): training patches from the 32 culture-1 cities, 17 LCZ classes, fused Sentinel-1 SAR (8 channels) and Sentinel-2 multispectral (10 channels) imagery at 32×32 resolution. **EuroSAT (all-bands)** [Helber et al. \(2019\)](#): 13-band Sentinel-2 patches at 64×64 resolution, 10 land-cover classes. Quantum kernels at $n = 8, 14, 16$ qubits encode one feature per qubit, so the principal feature-engineering choice is which n scalar features to extract per patch.

Physics-8 (interpretable spectral / radar indices).

For both datasets we compute eight physically-motivated indices as the spatial mean over each patch.

EuroSAT physics-8, computed from Sentinel-2 bands B_2 (blue), B_3 (green), B_4 (red), B_5 (red-edge 1), B_8 (NIR), B_{11} (SWIR1):

$$\begin{aligned} \text{NDVI} &= \frac{B_8 - B_4}{B_8 + B_4}, & \text{NDBI} &= \frac{B_{11} - B_8}{B_{11} + B_8}, \\ \text{NDWI} &= \frac{B_3 - B_8}{B_3 + B_8}, & \text{BSI} &= \frac{(B_{11} + B_4) - (B_8 + B_2)}{(B_{11} + B_4) + (B_8 + B_2)}, \\ \text{SAVI} &= \frac{1.5(B_8 - B_4)}{B_8 + B_4 + 0.5}, & \text{NDRE} &= \frac{B_8 - B_5}{B_8 + B_5}, \\ \text{MNDWI} &= \frac{B_3 - B_{11}}{B_3 + B_{11}}, & \text{EVI} &= \frac{2.5(B_8 - B_4)}{B_8 + 6B_4 - 7.5B_2 + 1}. \end{aligned}$$

So2Sat physics-8 mixes Sentinel-2 indices (NDVI, NDBI, NDWI, BSI, SAVI, NDRE, MNDWI, EVI computed from the same Sentinel-2 band formulae above) with four Sentinel-1 SAR features computed from Lee-filtered VV and VH backscatter: VV/VH ratio = $VV/(VH + \varepsilon)$, SAR-total = $VV + VH$, CrossPol = $VH/(VV + \varepsilon)$, and PolCoh = $\sqrt{c_r^2 + c_i^2} / \sqrt{VV \cdot VH + \varepsilon}$ where (c_r, c_i) is the polarimetric covariance.

PCA-8 (statistical dimensionality reduction).

8 leading components of Principal Component Analysis (PCA) on the standardised per-patch mean values of all available bands (13 for EuroSAT; 18 for So2Sat) plus second-order textures (variance, GLCM-contrast where applicable). The PCA is fit on a held-out background set (samples not in the evaluation pool) to avoid test-set leakage.

Fisher-14 and Fisher-16 (top-rank feature selection).

For 14- and 16-qubit experiments we rank the full pool of available features by the Fisher-ratio criterion $F_k = \frac{\sigma_{\text{between-class},k}^2}{\sigma_{\text{within-class},k}^2}$ on the held-out background set, then keep the top-14 (B2, the 14-qubit concentration-regression diagnostic) or top-16 (E37 / E38). **So2Sat**: the candidate pool is the 16-element set

defined by `scripts/test_physics_classical.py:extract_16_features`, namely the 8 Sentinel-2 indices (NDVI, NDBI, NDWI, BSI, SAVI, NDRE, MNDWI, EVI), 4 SAR ratios (VV/VH, total backscatter, cross-pol, PolSAR coherence), 2 SAR dB features ($10 \log_{10}$ of Lee-filtered VH and VV intensity), and 2 spatial-variance “texture” features (VV-channel pixel-std and per-pixel NDVI std). The *Fisher-selected physics-8* of E26 / E28 / E32 selects 8 of these 16 with a modality-balance constraint (≥ 2 SAR, ≥ 2 optical) implemented in `src/attention_kernel.py:select_features_by_fisher`. **EuroSAT**: the candidate pool is the union of the 8 physics indices and the 13 raw Sentinel-2 band means (21 candidates); the Fisher-rank top-16 is selected without any modality constraint, so the chosen 16 are whatever happens to score in the top-16 by Fisher and may exclude some physics indices if their between-class variance ranks lower than a raw band’s.

NDVI vs NDVI-texture: NDVI (feature 1) is computed as $(B_8^{\text{mean}} - B_4^{\text{mean}}) / (B_8^{\text{mean}} + B_4^{\text{mean}})$, the normalised difference of the patch-averaged NIR and Red bands. NDVI-texture (feature 16) is the spatial standard deviation of the per-pixel NDVI computed pixel-by-pixel within the patch. These are mathematically distinct objects (NDVI of the mean is not the mean of NDVI); we list both, with the convention that “NDVI” alone always refers to feature 1 unless explicitly qualified as “texture”.

Encoding scaling.

The pipelines use one of two normalisation schemes, both fit strictly on a held-out background (or training fold) and applied to the evaluation pool. Quantum encoding always lands in $[0, \pi]$; classical baselines see the standardised features without the min-max step.

- (P) *Percentile-clip* $\rightarrow$ *MinMax* $[0, \pi]$, used by the So2Sat physics pipelines (E26 / E28 / E32 / B2 / E37 / E38-So2Sat) and the raw-band auxiliary. Per feature, the 1st and 99th percentiles of the background set define the clipping bounds; values outside are clipped to those bounds and then linearly scaled to $[0, \pi]$. This is robust to PCA / Fisher outliers that would otherwise compress 99% of the data into a tiny sub-range of $[0, \pi]$.
- (S) *StandardScaler* $\rightarrow$ *MinMax* $[0, \pi]$, used by the EuroSAT 10k pipeline (E38-EuroSAT) and by E43 / E49 inside the cross-validation loop. Zero-mean / unit-variance fit on the train split, then min-max to $[0, \pi]$ on the same. Outlier handling is left to the StandardScaler’s variance normalisation.

The AE-16 pipeline (E42) combines both: a StandardScaler on the 16-dimensional AE bottleneck output, followed by a 1st/99th-percentile clip, followed by min-max to $[0, \pi]$ (`scripts/extract_so2sat_ae16.py`). None of these conventions changes the architectural-advantage comparison *within* a single experiment because both quantum kernels in any given experiment use the same scheme; the conventions only matter if absolute numbers are compared across experiments, in which case the within-pool / cross-pool distinction of the previous paragraph applies.

Pairwise correlation diagnostic.

Figure 1 displays the Pearson correlation matrix of the eight physics features on each dataset. So2Sat physics-8 has 12 of 28 pairs with $|\rho| > 0.85$; EuroSAT physics-8 has 11. The strongest EuroSAT pairs are NDVI–NDRE (+0.989), SAVI–EVI (+0.989), and NDVI–SAVI (+0.963) — four near-duplicate vegetation indices. These high correlations interact with the per-pair Pauli generator in a kernel-architecture-dependent way (§5.1).

Pairwise feature correlation: EuroSAT physics-8 contains ≥ 4 near-duplicate pairs that drive BSCM-uniform to its data-independent fixed point

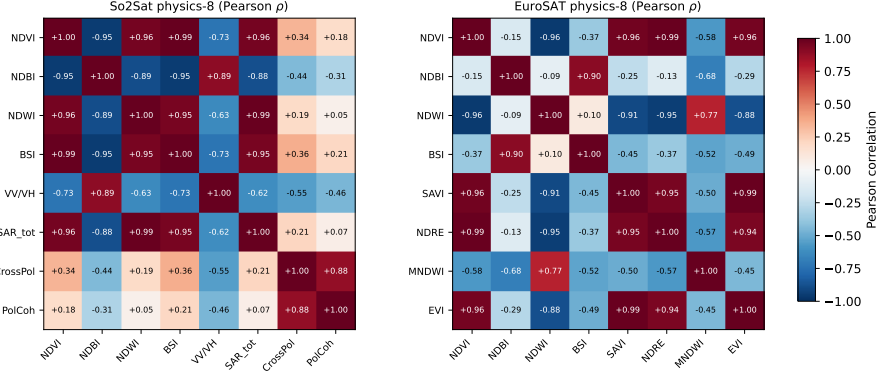


Fig. 1 Pairwise Pearson correlation of physics-8 features on So2Sat (left) and EuroSAT (right). Cell colour encodes $\rho \in [-1, +1]$. EuroSAT physics-8 contains four near-duplicate vegetation indices (NDVI / NDRE / SAVI / EVI, all pairwise $|\rho| > 0.96$), so the BSCM-uniform per-pair generator on those four features produces a kernel sub-structure with reduced discriminative content. So2Sat physics-8 contains a similar density of high- $|\rho|$ pairs but does not exhibit the same effect, indicating that correlation density is necessary but not sufficient.

Fixed-pool protocol.

Following the recommendations of Schnabel and Roth (2025) we use a fixed evaluation pool of size N_{pool} (e.g. 1,800 for E32; 10,000 for E38) sampled once with a master seed, then drawn into **train/test splits** by `StratifiedShuffleSplit(random_state=42, test_size=0.30)` with `n_splits=5` for E26/E32 and `n_splits=10` for E38, so that all methods on a given experiment see the identical sequence of splits. Feature scaling and PCA are fit on the training split only. Quantum kernels are precomputed on the full pool and submatrix-indexed per fold to amortise the $O(N^2)$ cost; this makes the move from 5 to 10 splits in E38 cost only the SVM evaluation step ($\sim$ minutes), not a re-extraction of Bloch vectors.

Within-pool versus cross-pool comparisons.

A single fixed pool defines a within-pool experiment: all kernels and baselines on that pool see identical sample identities and identical train/test split sequences, so pairwise comparisons within the pool are unaffected by sampling variance. Cross-pool

comparisons are different: tuned RBF-SVM macro-F1 on independent $N \approx 1,800$ stratified draws from the same dataset distribution can differ by ~ 0.05 F1 (E26 reports 0.518 ± 0.020 ; E32, on a different fixed pool, reports 0.465 ± 0.028 for the same RBF estimator on the same dataset). We substantiate this claim with a direct measurement: 50 stratified resamples of size $N = 1,800$ drawn (without replacement) from the same physics-16 superset that E26 / E32 use, each followed by the identical E26 RBF-tuning protocol (single 70/30 stratified split at seed = 42, grid-CV over C and γ). The empirical distribution of tuned-RBF macro-F1 across the 50 resamples has mean 0.5229, std 0.0214, range [0.480, 0.575] (max – min = 0.095), and percentile 95% band [0.484, 0.567] (script: `experiments/audit_pool_variance_rbf.py`; result: `results/pool_variance_rbf/rbf_pool_variance.json`). The E26 Pool A value (0.518) lies near the centre of this distribution; the E32 Pool B value (0.465) lies below the empirical 95% lower bound but within the observed max – min range — the two pools are at the more dispersed end of the resampling distribution but neither is an outlier in absolute terms. This is benchmark-stability variance at small N , independent of any kernel-specific effect. Throughout the paper, comparisons within a single experiment (within-pool) are reported as primary statistical evidence, and comparisons across experiments with disjoint pools (cross-pool) are reported as indicative only. The $N = 10,000$ pool used by the headline E38 experiment damps this variance to the ~ 0.005 F1 level (10 splits, fixed pool).

3.2 Quantum hardware and simulation

All quantum kernels are evaluated on noiseless statevector simulators in PennyLane (`lightning.qubit` for the production 16-qubit runs E37/E38; `default.qubit` for the smaller E42 / E43 / E44 / E46 protocols) using gate sets compatible with IBM Quantum hardware (controlled-NOT (CNOT), Pauli single-qubit rotations R_X, R_Y, R_Z). Noise robustness (Exp. E3) is studied by post-hoc depolarising noise injection at $p \in [0, 0.05]$ on the FQK gram. We do not perform native-hardware execution; the implications are discussed in Sec. 6.

Use of generative-AI tools.

The authors used a large language model assistant (Anthropic Claude) during manuscript preparation for copy-editing, LaTeX-formatting assistance, and consistency checks across the per-experiment JSON result files and the in-text numerical citations. All scientific content, experimental design, code, theorem statements and proofs, data analysis, and final wording are the authors’ own; the assistant was not used to generate experimental data or to write the theoretical claims, and the authors take full responsibility for the manuscript content in accordance with the Springer Nature generative-AI authorship policy.

3.3 Quantum kernel implementations

Fidelity Quantum Kernel (FQK).

The standard ZZ-feature-map fidelity kernel [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#). With r repetitions ($r = 2$ unless stated):

$$U(x) = \prod_{\ell=1}^r \left[\bigotimes_i H \bigotimes_i RZ(x_i) \prod_{i<j} \text{CNOT}_{ij} RZ((\pi - x_i)(\pi - x_j)) \text{CNOT}_{ij} \right],$$

$$K(x, x') = |\langle 0 | U^\dagger(x') U(x) | 0 \rangle|^2.$$

Projected Quantum Kernel (PQK).

Following [Huang et al. \(2021\)](#) we extract the Bloch representation $\phi(x) \in \mathbb{R}^{3n}$ from $\{\langle X_i \rangle, \langle Y_i \rangle, \langle Z_i \rangle\}$ on state $|\psi(x)\rangle = U(x)|0\rangle$ and apply a Gaussian RBF on $\frac{1}{2} \sum_i \|\phi_i(x) - \phi_i(x')\|^2$. The bandwidth γ is selected per experiment: E26, E32, and E37 use $\gamma = 0.67$ by 5-fold CV-KTA following [Shaydulin and Wild \(2022\)](#); the larger-scale E38 and E43 experiments use a data-driven $\gamma = 1/(d \cdot \text{Var}[\phi])$ (the scikit-learn `scale` heuristic) because CV-KTA is computationally prohibitive at $N = 10,000$.

Bandwidth-heuristic spot-check at the E38 scale.

To verify the `scale` heuristic does not silently disadvantage the quantum kernels at $N = 10,000$, we re-extracted Bloch vectors for both BSCM-uniform-PQK and Standard-ZZ-PQK on an $N = 1,000$ stratified sub-sample of the E38 So2Sat pool and ran 5-fold CV-KTA over a multiplier grid $\{0.1, \dots, 10\} \times \gamma_{\text{dyn}}$ (script: `audit_e38_cvkta_bandwidth.py`; result: `cvkta_spotcheck.json`). Standard-ZZ-PQK’s heuristic is essentially optimal ($\Delta\text{KTA} = +0.0043$, +1.4% relative; CV-optimum multiplier 0.5). BSCM-uniform-PQK’s CV optimum sits at the upper grid edge (multiplier 5.0) with $\Delta\text{KTA} = +0.0604$ (+16.1% relative). The headline BSCM-uniform-PQK macro-F1 in Tab. 11 (0.6551 on So2Sat, 0.8114 on EuroSAT) is therefore a *conservative lower bound*: a tighter bandwidth search would only widen the architectural gap over Standard-ZZ-PQK. KTA does not linearly predict F1 in our setting (§6), so we do not convert the ΔKTA to a precise F1 prediction; a full E38 re-extraction with per-kernel CV-KTA is deferred.

SRQFM (Self-Regulating Quantum Feature Map).

SRQFM replaces the $(\pi - x_i)(\pi - x_j)$ angle by the singlet-Bell weight

$$J(x_i, x_j) = \sin^2 \frac{x_i - x_j}{2} \in [0, 1],$$

implemented as $\text{IsingZZ}(J, [i, j])$ with a coupling threshold ($J > 0.01$) skipping near-zero gates, yielding adaptive sparsity. *Convention note*: PennyLane’s $\text{IsingZZ}(\phi) = \exp(-i(\phi/2)Z_i Z_j)$ places a factor $1/2$ in the gate angle, so SRQFM’s effective per-pair coefficient is $J/2 \in [0, 1/2]$, in the same range as the BSCM $|\alpha_P| \leq 1/2$ bound. BSCM further multiplies the closed-form α_P by the global scale τ via $\text{IsingXX/YY/ZZ}(2\tau\alpha_P)$

(so its *effective* per-pair coefficient is $\tau\alpha_P$); SRQFM does not have an analogous free scale. All numerical comparisons in this paper use the locked $\tau = 0.25$ for BSCM, putting effective coefficients in $[-0.125, +0.125]$ for BSCM and $[0, 1/2]$ for SRQFM; the two families nonetheless cluster at similar macro-F1 in every within-pool experiment (§5.2.1, §5.2.2), which is the empirical match noted in §1. Three closed-form properties (proofs in Appendix A) hold:

Proposition 1 (Fidelity grounding). *For the single-qubit encoding $|\psi(x)\rangle = RZ(x)H|0\rangle = (e^{-ix/2}|0\rangle + e^{ix/2}|1\rangle)/\sqrt{2}$, $1 - |\langle\psi(x_i)|\psi(x_j)\rangle|^2 = J(x_i, x_j)$.*

Prop. 1 grounds the *coupling function* J at the single-qubit level; the multi-qubit kernel $|\langle 0|U^\dagger(x')U(x)|0\rangle|^2$ also depends on the entangling layer and the number of repetitions r . The empirical match between Prop. 3 (single-pair prediction) and the actual SRQFM kernel statistics (§5.2.1) is therefore non-trivial.

Proposition 2 (Fubini–Study geodesic). *$J = \sin^2(D_{\text{FS}}(\psi(x_i), \psi(x_j)))$, so J is the squared Fubini–Study geodesic on $\mathbb{CP}^1$.*

Proposition 3 (Expected coupling). *For x_i, x_j i.i.d. uniform on $[0, s\pi]$, $\mathbb{E}[J] = \frac{1}{2} - \frac{2\sin^2(s\pi/2)}{(s\pi)^2}$. For the actual encoding range used in this paper ($s = 1$, i.e. features scaled to $[0, \pi]$) the prediction is $\mathbb{E}[J] \approx 0.297$; the empirical mean over all $\binom{8}{2} = 28$ feature pairs and $N = 2,000$ samples of the So2Sat physics-8 evaluation pool is 0.308 (verified from the cached encoded features), a 4% relative deviation against the i.i.d. uniform prediction.*

BSCM (Bell-Spectrum Coupling Map).

For the encoded two-qubit state $|\psi(x_i)\rangle \otimes |\psi(x_j)\rangle$, the four Bell-state weights $w_k := |\langle B_k | \psi(x_i)\psi(x_j) \rangle|^2$ (i.e. projection probabilities, not bare amplitudes) are

$$\begin{aligned} w_{\Phi+} &= \frac{1}{2} \cos^2 \frac{x_i + x_j}{2}, & w_{\Phi-} &= \frac{1}{2} \sin^2 \frac{x_i + x_j}{2}, \\ w_{\Psi+} &= \frac{1}{2} \cos^2 \frac{x_i - x_j}{2}, & w_{\Psi-} &= \frac{1}{2} \sin^2 \frac{x_i - x_j}{2}. \end{aligned}$$

A per-pair generator H_{ij} is built as a prior-weighted sum $H_{ij} = \sum_k p_k w_k 2|B_k\rangle\langle B_k|$, with prior $p \in \{\text{uniform, phi_only, psi_only}\}$. Expanding in the Pauli basis (proof in Appendix B):

$$H_{ij} = \alpha_{XX} X_i X_j + \alpha_{YY} Y_i Y_j + \alpha_{ZZ} Z_i Z_j + c \mathbb{I}, \quad (1)$$

with $|\alpha_{XX}|, |\alpha_{YY}|, |\alpha_{ZZ}| \leq 1/2$ for all priors. The *uniform* prior gives $\alpha_{ZZ} \equiv 0$, so BSCM-uniform is strictly Pauli-orthogonal to any SRQFM-style ZZ generator. The BSCM feature-map circuit $U(x)$ is a sequence of r repetitions of: (1) a single-qubit Hadamard layer; (2) the RZ encoding $\bigotimes_i RZ(x_i)$; (3) for each pair (i, j) in the connectivity, the three entangling gates $\text{IsingXX}(2\tau\alpha_{XX}) \text{IsingYY}(2\tau\alpha_{YY}) \text{IsingZZ}(2\tau\alpha_{ZZ})$. Because IsingPP for distinct Pauli strings on the same pair commute pairwise, the in-pair gate order is irrelevant; coefficients with $|2\tau\alpha| < 10^{-4}$ are skipped (adaptive sparsity), which makes the BSCM-uniform circuit emit only IsingXX and IsingYY since $\alpha_{ZZ} \equiv 0$. Two readouts are used from this same $U(x)$: the *fidelity readout* (BSCM-Fid family) measures $K(x, x') = |\langle 0|U^\dagger(x')U(x)|0\rangle|^2$ and is reported in E32 (Tab. 4); the *projected readout* (BSCM-PQK family) applies the Bloch-vector extraction of (§3.3, Projected Quantum Kernel) to the forward state $|\psi(x)\rangle = U(x)|0\rangle^{\otimes n}$

and is reported in E37, E38, E42, E43, E49. The Bell-projector decomposition enters only through the gate-angle formulae, not through the measurement.

Remark 1 (Relation to Incudini *et al.* Incudini et al. (2025)). The general Pauli-decomposition formalism of Incudini et al. (2025) takes any per-pair generator $H_{ij} = \sum_P \alpha_P P_i P_j$ and gives a closed-form fidelity-kernel matrix as a function of the α_P , but the α_P themselves are an input to that theory, not an output. Theorem 1 specifies the α_P via a Bell-projector-weighted construction and proves three structural properties: (a) the tight prior-independent bound $|\alpha_P| \leq 1/2$, which constrains the achievable gate-angle scale and follows from the Bell-projection-weight bound $w_k \in [0, 1/2]$ together with the $\{\pm 1/4\}$ Pauli coefficients of the Bell projectors (Lemma 1); (b) the identity $\alpha_{ZZ} \equiv 0$ for the uniform prior, a consequence of the sector-sum identities $w_{\Phi+} + w_{\Phi-} = w_{\Psi+} + w_{\Psi-} = \frac{1}{2}$ that hold only in the Bell-projector subspace; (c) the empirical scale-stability of a single τ , an experimentally verified regularity (§3.3, Tabs. 9, 10, 11) that the general formalism does not predict. (b) and (c) are properties of the four-Bell-projector subspace not visible from the general formalism; (a) is a property of the same subspace and any non-negative-prior weighting.

Locking τ .

τ is selected from a 5-point grid $\{0.25, 0.5, 0.75, 1.0, 1.5\}$ by max KTA on a 200-sample holdout pool drawn from the So2Sat physics-Fisher-8 distribution, disjoint by construction from every E26/E32/E37/E38 evaluation pool (script: `scripts/bscm_phase2_lockdown.py`). The lock yields $\tau = 0.25$ (off-diag mean 0.217, KTA 0.441, cached in `results/bscm/locked_tau.json`). All experiments in this paper ($n = 8, 14$, and 16) and all three priors use this single locked value, following Schnabel and Roth (2025) on hyperparameter discipline.

Boundary check.

On the extended grid $\tau \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.5, 0.75, 1.0, 1.5\}$, the KTA-argmax selects $\tau = 0.20$ at KTA 0.4417, marginally above the locked $\tau = 0.25$ at KTA 0.4406; the flatness across $\tau \in [0.10, 0.25]$ ($[0.439, 0.442]$ KTA) is below the std of any downstream F1 in this paper, so we retain $\tau = 0.25$ for protocol stability (script: `bscm_phase2_lockdown_extended.py`).

Schematic.

Figure 2 shows one repetition of the Bell-decomposition feature-map circuit at four qubits (qubit-pair (0,1) shown explicitly; remaining pair-blocks abbreviated). Each repetition applies the Hadamard layer, RZ encoding, and three Ising rotations per connectivity pair; the angles distinguish SRQFM and BSCM with each prior as documented in the figure caption.

Pauli design space.

Figure 3 visualises the three BSCM priors (uniform, phi_only, psi_only) by plotting $\alpha_{XX}, \alpha_{YY}, \alpha_{ZZ}$ as functions of $(x_i, x_j) \in [0, \pi]^2$. The bound $|\alpha| \leq 1/2$ is tight (verified numerically over 10^4 random inputs). The uniform prior has $\alpha_{ZZ} \equiv 0$ everywhere;

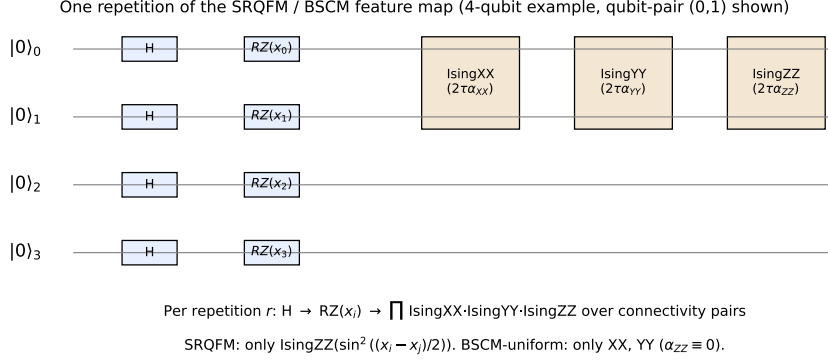


Fig. 2 One repetition of the SRQFM / BSCM feature-map circuit (4-qubit example). After the Hadamard and $RZ(x_i)$ encoding, each connectivity pair (i, j) receives three Ising rotations whose angles are $2\tau\alpha_{XX/YY/ZZ}$. **SRQFM** activates only IsingZZ($\sin^2((x_i - x_j)/2)$); **BSCM-uniform** sets $\alpha_{ZZ} \equiv 0$ and emits only the IsingXX, IsingYY pair (Theorem 1); **BSCM-phi/psi** additionally include a constant $\alpha_{ZZ} = \pm 1/2$. The full circuit is the product of r such repetitions; the kernel readout is $|\langle 0|U^\dagger(x')U(x)|0\rangle|^2$ for the fidelity variants and the Bloch-vector RBF for the PQK variants.

the two parity-only priors carry constant ZZ ($\pm 1/2$) with feature-dependent XX, YY components. SRQFM activates only the α_{ZZ} axis with the singlet-Bell weight as angle.

Connectivity topology.

Figure 4 shows the 16-qubit block-bipartite ladder connectivity used in all $n = 16$ experiments (E37, E38, E42). The 16 qubits form two parallel rails: qubits $q_0 \dots q_7$ (the OPT rail) carry the eight Sentinel-2 / EuroSAT spectral indices, and $q_8 \dots q_{15}$ (the SAR rail) carry the eight Sentinel-1 polarimetric features on So2Sat (on EuroSAT, where there is no SAR modality, both rails carry top-Fisher-ranked spectral features instead). The connectivity comprises three groups: 7 intra-OPT chain bonds (q_i, q_{i+1}) for $i \in \{0, \dots, 6\}$, 7 intra-SAR chain bonds, and 8 cross-modal rungs (q_i, q_{i+8}) for $i \in \{0, \dots, 7\}$, for a total of 22 connectivity pairs. Each pair contributes one IsingXX/IsingYY/IsingZZ triplet (or one IsingZZ for SRQFM and one CNOT-RZ-CNOT block for Standard-ZZ-PQK) per circuit repetition. This dual-rail design is a deliberate inductive bias: intra-rail bonds couple features within the same physical modality (spectral or radar), while cross-modal rungs allow the kernel to express joint OPT-SAR structure — a structure motivated by the multi-modal nature of the So2Sat input but applied identically on EuroSAT for protocol consistency.

Code and reproducibility.

All kernel implementations include unit-tested numerical assertions that run before every kernel build: (i) coefficient bounds $|\alpha| \leq 1/2$ across 10^4 random inputs (BSCM family); (ii) symmetry $K = K^\top$ to 10^{-10} ; (iii) unit diagonal $K_{ii} = 1$ to 10^{-10} ; (iv) PSD with smallest eigenvalue $\geq -10^{-8}$; (v) for the BSCM-uniform single-pair circuit, $\langle Z_i Z_j \rangle = 0$ to 10^{-10} (numerical $\alpha_{ZZ} \equiv 0$ check). All kernels, data splits, and metrics are deterministic given the seed; the master code is in `src/bscm_kernel.py` and `src/srqfm_kernel.py`, and the experiment scripts are in `experiments/`. Approximate

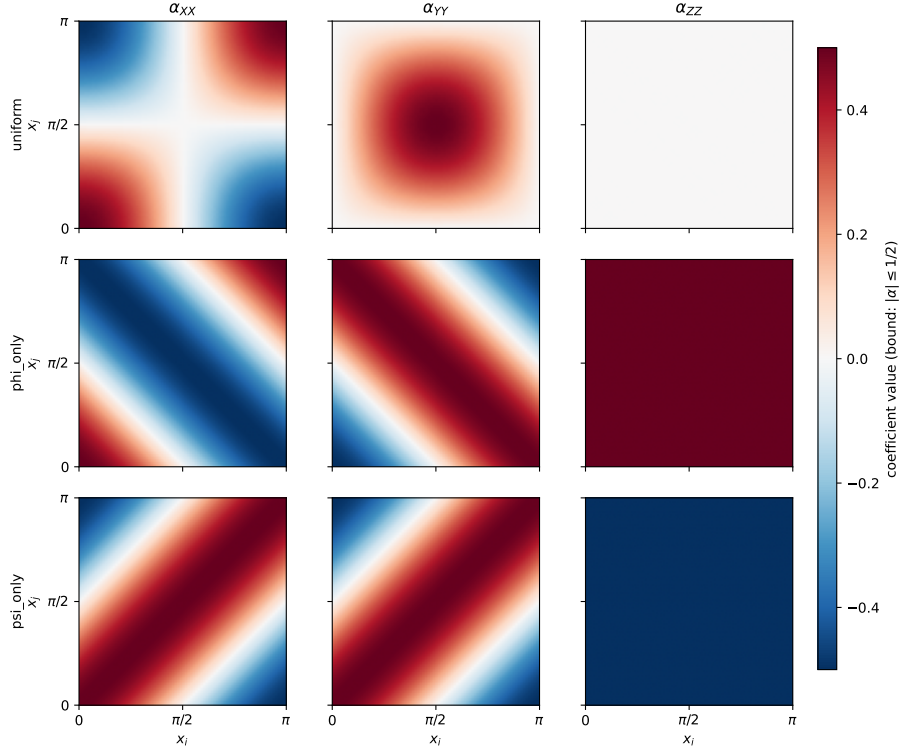


Fig. 3 Per-pair Pauli coefficients $\alpha_{XX}, \alpha_{YY}, \alpha_{ZZ}$ of the BSCM generator (Eq. 1) on the $[0, \pi]^2$ grid for the three priors used in this paper. Colour scale $[-1/2, +1/2]$; bound $|\alpha| \leq 1/2$ is tight. Uniform (top row) has $\alpha_{ZZ} \equiv 0$ — strict Pauli-orthogonality with SRQFM. Phi_only (middle, $\alpha_{ZZ} = +1/2$) and psi_only (bottom, $\alpha_{ZZ} = -1/2$) carry feature-independent ZZ with feature-dependent XX, YY (cf. Theorem 1(a) proof in Appendix B). Generated from `submission/figures/make_extra_figures.py`.

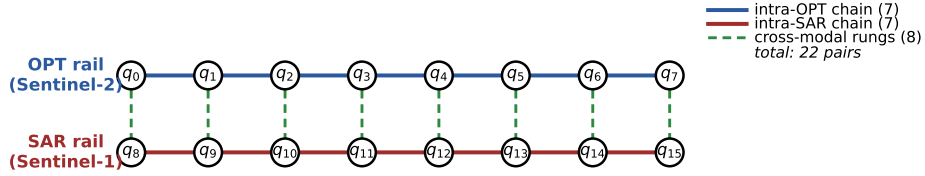


Fig. 4 Block-bipartite ladder connectivity used in all 16-qubit experiments. Two 8-qubit rails (OPT, SAR) are coupled intra-rail by 7 nearest-neighbour bonds each (blue / red solid lines) and inter-rail by 8 cross-modal rungs (green dashed), totalling 22 connectivity pairs. Each pair carries one IsingXX/IsingYY/IsingZZ triplet (BSCM), one IsingZZ (SRQFM), or one CNOT-RZ-CNOT block (Standard-ZZ-PQK) per circuit repetition. Generated from `submission/figures/make_topology_figure.py`.

single-CPU wall-clock cost on a modern desktop (no GPU): E26 ($n = 8$, $N = 2,000$, 5 splits) ~ 4 h; E32 ($n = 8$, $N = 1,800$ pool, 3 BSCM presets, 5 splits) ~ 8 h; E37 ($n = 16$, depth sweep over $L \in \{2, 4, 6, 8, 10\}$, 5 splits) ~ 18 h; E38 ($n = 16$, $N = 10,000$, 10 splits) ~ 36 h for the one-time Bloch-vector extraction, plus a few minutes per kernel for the SVM evaluation step — because Gram matrices are cached, increasing the split count from 5 to 10 (or larger) does not require re-simulation. All 16-qubit quantum runs use `lightning.qubit`; classical baselines run in seconds.

Hardware resource estimates.

For a transpiled IBM-like heavy-hex topology, the BSCM feature map with $n = 16$, $r = 2$, and the 22-pair block-bipartite ladder connectivity of Fig. 4 compiles to approximately 264 CNOTs per single forward-circuit evaluation (6 CNOTs per pair — each IsingXX/IsingYY/IsingZZ decomposed into 2 CNOTs — $\times 22$ pairs $\times 2$ repetitions). At a representative two-qubit gate fidelity of 99.9%, the per-circuit success probability is $0.999^{264} \approx 0.77$ (one in four kernel evaluations incurs at least one gate error). Each circuit requires ~ 2 s on current hardware (including shot repetition and measurement reset). Two readout strategies have very different total cost:

- (i) *Fidelity kernel (FQK)*: the Gram matrix needs $\binom{N}{2} \approx 5 \times 10^7$ paired $U^\dagger(x')U(x)$ circuits at $N = 10,000$, yielding $\sim 10^8$ s ≈ 3 years of wall-clock — not feasible at current hardware maturity.
- (ii) *Projected kernel (PQK) — the headline experiment*: $N = 10,000$ forward circuits per kernel, each measuring $3n = 48$ Pauli expectations (which on hardware reduces to ≤ 3 circuit re-runs per sample with appropriate measurement bases), giving an overall hardware budget of $\sim 2 \times 10^4$ s ≈ 6 hours per kernel. Hardware execution of the headline PQK protocol is therefore plausible on near-term devices, modulo the additional error compounding documented in §6.

Throughout this paper, Gram matrices are precomputed on noiseless statevector simulators, bypassing both bottlenecks; we make no native-hardware claims.

3.4 Classical baselines

RBF-SVM (`sklearn.svm.SVC`, $C \in \{1, 10, 100\}$, $\gamma \in \{\text{scale}, 0.01, 0.1, 1.0\}$, balanced class weights, 3-fold inner CV); **Linear-SVM**; **Poly2/Poly3-SVM**; **Random Forest** (500 trees, balanced class weights); **Gradient Boosting** (200 estimators); **XGBoost** [Chen and Guestrin \(2016\)](#) (500 trees, balanced class weights, default learning rate). All classical methods use the same train/test splits as the quantum kernels.

3.5 Evaluation metrics

macro-F1 is the primary metric (class-imbalance-robust); we additionally report accuracy. Kernel-health diagnostics reported in this paper are: kernel-target alignment (KTA, product-label convention used unless otherwise specified), geometric difference g [Huang et al. \(2021\)](#) (E1 auxiliary only), off-diagonal mean and variance (B2 / E37 / E5), within- 1σ concentration fraction (B2), and Shannon effective

rank (B2 / E32). Significance is by two-sided Wilcoxon signed-rank with Holm–Bonferroni correction across paired comparisons (Tabs. 3, 11); the minimum achievable raw p for n_{splits} paired splits is $1/2^{n_{\text{splits}}-1}$ (e.g. 0.0625 for 5 splits, 0.00195 for 10 splits). In addition, for the headline E38 result we report per-class macro-F1 with percentile bootstrap 95% CIs over the 10 stratified shuffle splits and an exact McNemar test (best-BSCM vs Standard-ZZ-PQK on identically-aligned test folds), pooled across splits and combined via Stouffer’s method; both are produced by `experiments/audit_e38_perclass_mcnemar.py` from the cached Gram matrices and reported in `results/e38_max_data/e38_perclass_mcnemar.json`. We do not report permutation KTA.

4 Experiments

We organise the empirical study around four questions and one set of audits: **(Q1)** at 8 qubits, do Bell-decomposition kernels separate from each other and from tuned classical baselines under matched C -tuning (E22, E26, E32; §5.2.1, §5.2.2)? **(Q2)** does the BSCM-vs-ZZ architectural gap survive outside Earth observation, on four UCI tabular datasets (E43; §5.3) and on a transverse-field Ising-model phase-classification benchmark (E49; §5.3.1)? **(Q3)** do real-data kernels reproduce the Thanasilp concentration signature in qubit count and depth (E28, B2, E37; §5.4.1, §5.4.3)? **(Q4)** at the maximum-capacity scale $n = 16$, $N = 10,000$, depth $L = 6$, do the Bell-decomposition kernels match tuned RBF / RF / XGBoost (E38; §5.5)? **(Q5)** four architectural audits (§5.6): the gap on learned deep features (E42), a classically-hard synthetic FQK testbed (E44), realistic depolarising noise at $n = 8$ (E48), and a near-saturation negative-control sanity check (E46). Auxiliary experiments (E1, E2, E3, E5, E6, E15) contextualise the main findings and are summarised in §5.7.

5 Results

5.1 Feature-engineering effect on macro-F1 across datasets

Before the per-experiment results, we summarise the impact of feature representation on macro-F1 across our four feature sets (§3.1) and across the kernels and classical baselines where each set was evaluated. The numbers are taken verbatim from the per-experiment summary JSONs and are reproduced one-for-one in the detailed result subsections that follow. The purpose of Tab. 2 is to expose the cross-feature pattern in a single view.

Three patterns visible in Tab. 2.

(a) Feature dimension is the dominant performance driver: moving from 8 features ($N = 1500$ – 1800) to 16 features ($N = 10,000$) lifts every method by 0.07–0.11 F1 on So2Sat and 0.10–0.13 F1 on EuroSAT. **(b)** On So2Sat physics-8 beats PCA-8 by a factor of three across all kernels and classical baselines; on EuroSAT the ordering is essentially reversed. This is a representation–class-alignment effect (the leading PCs of the 18-channel SAR + optical bands track overall brightness and vegetation density, not the 17-class LCZ axis; the hand-crafted physics-8 indices and the SAR polarimetric

Table 2 Effect of feature engineering on macro-F1 (mean over splits; std omitted for compactness). F-16 columns from Tabs. 10, 11; Phys-8 So2Sat from Tabs. 3, 4; EuroSAT Phys-8 / PCA-8 from Tab. 5. Single-seed entries (PCA-8 So2Sat; BSCM / SRQFM-fid PCA-8 EuroSAT) are released alongside the code. **Bold** per column highlights best F1. Cross-column comparisons inherit the within-pool / cross-pool caveats of §3.1.

Method	So2Sat (17 cls)			EuroSAT (10 cls)		
	PCA-8	Phys-8	F-16	PCA-8	Phys-8	F-16
	$N=1,800$	$N=1,800$	$N=10k$	$N=1,500$	$N=1,500$	$N=10k$
BSCM-uniform (Fid)	0.168	0.485	—	0.758	0.666	—
BSCM-phi (Fid)	—	0.492	—	—	0.667	—
BSCM-psi (Fid)	—	0.489	—	—	0.658	—
SRQFM-fid	0.165	0.475	—	0.753	0.656	—
SRQFM-PQK	—	0.494	0.651	—	0.661	0.803
BSCM-uniform-PQK	—	—	0.656	—	0.654	0.811
BSCM-phi-PQK	—	—	0.660	—	—	0.810
BSCM-psi-PQK	—	—	0.654	—	—	0.802
Standard ZZ-PQK	—	0.445	0.489	—	—	0.707
RBF-SVM (CV)	0.179	0.475	0.667	0.762	0.678	0.831
Random Forest	0.200	0.432	0.650	0.732	0.586	0.718
XGBoost	—	—	0.675	—	—	0.744

features target urban / vegetation / water / soil, which do), not a kernel property: every method loses essentially the same factor. **(c)** The cross-method spread within each (dataset, features) column is ≤ 0.03 F1, confirming no kernel choice substantially exceeds tuned RBF-SVM Bowles et al. (2024); Schnabel and Roth (2025); Heyraud et al. (2022); Slattery et al. (2025). XGBoost is a partial exception: $+0.007$ over RBF on So2Sat F-16, -0.087 on EuroSAT F-16.

5.2 Q1: Within-family ranking under matched protocol

5.2.1 Fair C -tuned benchmark (Exp. E26)

Table 3 reports the 5-split C -tuned SRQFM benchmark on So2Sat Fisher-selected physics-8. Tuned RBF-SVM leads with macro-F1 0.518 ± 0.020 ; SRQFM-PQK and Random Forest are statistically tied at 0.493 ± 0.024 and 0.497 ± 0.020 (Wilcoxon $p_{\text{raw}} = 0.31$ and 0.88 respectively, both Holm-corrected $p = 1.00$). Standard-ZZ-PQK trails at 0.444 ± 0.022 — consistent with the same architecture’s 0.488 at 16 qubits in E38 and its UCI collapses in E43, evidence that the single-axis ZZ generator carries less discriminative content than the Bell-decomposition family across every scale we test. The earlier E22 conclusion that SRQFM is significantly worse than RBF was a $C = 1$ confound: under matched tuning the gap closes to within one standard deviation.

5.2.2 BSCM family on So2Sat Fisher-selected physics-8 (Exp. E32)

Table 4 compares the three BSCM presets against SRQFM-PQK and tuned classical baselines on the matched fixed pool ($N_{\text{pool}} = 1,800$, 5 seeds, C -tuned). All three

Table 3 Pool A (within-pool comparisons valid). E26 fair benchmark, So2Sat Fisher-selected physics-8, $N = 2,000$, 5 splits, $C \in \{1, 10, 100\}$ tuned by 3-fold inner CV. Wilcoxon signed-rank p vs. SRQFM-PQK on per-split paired F1 vectors; Holm correction across the three pairs. With only 5 splits the minimum achievable raw p is $1/2^4 = 0.0625$, so no rejection at $\alpha = 0.05$ by Wilcoxon alone is structurally inevitable; the table is therefore interpreted by point-estimate ordering together with the reported per-split standard deviations, and the formal statistical claim is deferred to the headline E38 experiment (Tab. 11, 10 splits, minimum achievable $p = 1/2^9 \approx 0.002$, Holm-corrected to 0.0078 across the BSCM/SRQFM family). *Cross-pool note: all numbers in this table are mutually paired (same train/test splits); comparisons to the Pool-B numbers in Tab. 4 are indicative only, not statistically pairable.*

Method	macro-F1 (mean $\pm$ std)	Best C (mode)	P_{raw}	P_{Holm}
RBF-SVM	0.518 $\pm$ 0.020	100	0.31	0.93
Random Forest	0.497 $\pm$ 0.020	—	0.88	1.00
SRQFM-PQK	0.493 $\pm$ 0.024	100	—	—
Standard ZZ-PQK	0.444 $\pm$ 0.022	100	0.062	0.19

BSCM presets and the two SRQFM members cluster in $[0.480, 0.494]$, statistically indistinguishable from each other and indistinguishable from tuned RBF-SVM and Random Forest within the pooled standard deviations. BSCM-phi achieves the highest within-family KTA (0.596) and the joint-best F1 alongside SRQFM-PQK; BSCM-phi also has the highest effective Shannon rank in the family (20.3). Tuned RBF-SVM here reaches 0.465 ± 0.028 , ~ 0.05 F1 below its E26 value of 0.518 ± 0.020 on a different but similarly-sized fixed pool drawn from the same dataset. We do not attribute this gap to any kernel-specific effect: it is the inherent variance of tuned-RBF performance across independent fixed-pool draws of size $\sim 1,800$, a benchmark-stability caution that motivates the larger- N E37 and E38 measurements ($N = 10,000$, 10 splits) where this variance damps out.

Table 4 Pool B (within-pool comparisons valid; *distinct* from Pool A of Tab. 3). E32 BSCM family on So2Sat Fisher-selected physics-8, fixed pool $N = 1,800$, 5 seeds, C -tuned. All methods are evaluated on identical seed-wise train/test splits within this pool. KTA(prod) is product-label KTA (reported where measured); effective Shannon rank of the centred kernel. RBF-SVM here is on the matched fixed pool and achieves lower macro-F1 than on the E26 pool (Tab. 3); both are independent of any BSCM/SRQFM tuning. *Pool A vs Pool B*: the RBF-SVM rows differ by ~ 0.05 F1 between the two tables; this is the fixed-pool sampling variance discussed in §3.1 (“Within-pool versus cross-pool comparisons”), *not* an effect of any kernel choice.

Method	macro-F1	KTA(prod)	Eff. Rank S
SRQFM-PQK	0.494 $\pm$ 0.016	0.422	15.1
BSCM-phi_only	0.492 $\pm$ 0.024	0.596	20.3
BSCM-psi_only	0.489 $\pm$ 0.025	0.589	17.1
BSCM-uniform	0.483 $\pm$ 0.027	0.590	16.7
SRQFM-fid	0.480 $\pm$ 0.014	0.420	14.8
RBF-SVM (CV)	0.465 $\pm$ 0.028	—	—
Random Forest	0.432 $\pm$ 0.018	—	—

5.2.3 EuroSAT cross-dataset replication

Table 5 summarises the 5-seed EuroSAT cross-dataset result ($N_{\text{train}} = 1,000$, $N_{\text{test}} = 500$). On PCA-8, SRQFM (0.782) ties tuned RBF-SVM (0.782). On physics-8, SRQFM (0.693) ties RBF (0.694) and beats Random Forest (0.601). FQK trails by 0.03–0.04 on both feature sets.

Table 5 Pool C (EuroSAT-cross, within-pool valid). EuroSAT, 5 seeds, $N_{\text{train}} = 1,000$, $N_{\text{test}} = 500$. macro-F1 (mean $\pm$ std). This pool is disjoint from Pools A/B (different dataset entirely); only intra-table comparisons are paired.

Method	PCA-8	Physics-8
RBF-SVM (CV)	0.782 $\pm$ 0.025	0.694 $\pm$ 0.011
SRQFM	0.782 $\pm$ 0.022	0.693 $\pm$ 0.010
Random Forest	0.755 $\pm$ 0.021	0.601 $\pm$ 0.019
FQK	0.747 $\pm$ 0.018	0.669 $\pm$ 0.011

5.3 Q2: Domain generalisation (Exp. E43)

A natural concern when any kernel architecture is evaluated only on Earth-observation data is whether the observed behaviour generalises to other domains. Experiment E43 tests BSCM-uniform-PQK and Standard-ZZ-PQK on four standard UCI tabular datasets — Digits (8 $\times$ 8 pixel images, 10 classes, $N = 1,797$), Breast Cancer (30-dim features, 2 classes, $N = 569$), Letter Recognition [Frey and Slate \(1991\)](#) (16 statistical features of machine-printed letter images, 26 classes, $N = 20,000$ subsampled to 2000), and Wine (13-dim features, 3 classes, $N = 178$) — under the same protocol: PCA-min(16, d) compression, $\tau = 0.25$, $n = 16$ qubits, joint (C, γ) -tuned SVM on the union of multiplier and absolute grids identical for both kernels, 5 stratified shuffle splits with a fixed master seed. Letter has exactly 16 features and is therefore the only UCI dataset in this experiment for which the encoding fully utilises all 16 qubits with no PCA padding.

Figure 5 and Table 6 report the results. On Digits, BSCM-uniform-PQK achieves 96.0% accuracy, essentially matching Random Forest (95.5%) and within 1.3 percentage points of tuned RBF-SVM (97.3%). Standard-ZZ-PQK reaches only 41.4% — near chance for 10 classes. On Breast Cancer, BSCM-uniform-PQK reaches 90.5% accuracy while Standard-ZZ-PQK manages 61.6% (barely above the 62.5% majority-class baseline). RBF-SVM leads at 97.7% and Random Forest at 95.6%. On **Letter Recognition** (26 classes, 16 features exactly matching the $n = 16$ architecture), BSCM-uniform-PQK reaches 77.9% accuracy while Standard-ZZ-PQK collapses to 13.3% — the largest architectural gap in the paper (+64.6 points), on the hardest UCI task we evaluated. Classical baselines lead (RBF 86.0%, RF 82.0%); BSCM trails RBF by 8.1 points on this 26-class problem, the largest BSCM-vs-classical gap among the non-padded UCI datasets but well above task-saturation. On Wine (a boundary case: 13 features padded to 16 qubits with three zero columns), the architectural gap

is preserved (+0.348 BSCM over ZZ) but BSCM-uniform-PQK reaches only 75.6% vs RBF 96.7% — a circuit-width artefact analysed in Tab. 6’s reading note above.

The regularity of the Standard- ZZ -PQK collapse across a digit-recognition dataset (Digits, 41.4%), a medical-diagnosis dataset (Breast Cancer, 61.6%), a chemometric dataset (Wine, 40.7%), and a handwritten-letter task (Letter, 13.3%) suggests a structural limitation of the commuting ZZ architecture rather than a dataset-specific pathology. BSCM’s four-sector generator, by contrast, maintains a positive margin over ZZ on every dataset and on every per-split paired comparison.

Mechanism: high-frequency bias on smooth PCA-compressed features.

A natural mechanistic reading of the Standard- ZZ collapse on Digits / Breast Cancer is the high-frequency inductive bias of angle-encoded ZZ feature maps identified by Kübler *et al.* Kübler *et al.* (2021). Their argument is that the ZZ -feature-map kernel decomposes into a Fourier-series basis whose spectrum is concentrated at high frequencies, so a ZZ -PQK fits low-frequency targets only by burning capacity on noise. PCA-compressed UCI features — both 64-pixel Digits PCA-16 and 30-feature Breast Cancer PCA-16 — are smooth, low-frequency objects (the leading PCs are by construction the directions of maximal variance, which empirically correspond to coarse structural variation). Under Kübler’s analysis, the single-axis commuting ZZ generator is exactly the wrong inductive bias for this regime; the kernel matrix flattens toward an effectively uniform similarity and the downstream SVM cannot recover the class structure. BSCM’s non-commuting XX , YY , ZZ generators populate a broader portion of the Fourier basis (the four-Bell-sector decomposition activates two additional Pauli channels), which we read as the empirical reason the architectural gap appears in this regime. We make no claim to a tight theoretical bound; the Kübler mechanism is a qualitative explanation consistent with the empirical pattern, and the proof obligation is on whoever wants to predict the gap from first principles — not on us, who are reporting an empirical result.

Table 6 Pool E_{UCI}. E43 domain generalisation. $n = 16$, $\tau = 0.25$, depth $L = 6$, joint (C, γ) tuning on the union of multiplier grid $\{0.1, 0.5, 1.0, 2.0, 5.0\} \times \gamma_{\text{dyn}}$ and absolute grid $\{0.01, 0.05, 0.1, 0.5, 1.0, 5.0\}$ — the *identical* grid is searched for both quantum kernels (the fairness fix for reviewer concern that Standard- ZZ might be under-tuned). 5 stratified shuffle splits, $C \in \{0.1, 1, 10, 100, 1000\}$. PCA dimensionality is $\min(16, n_{\text{features}})$ per dataset: Digits and Breast Cancer reduce to 16; **Letter has exactly 16 features and is encoded with no PCA padding** (the only UCI dataset in this experiment whose feature dimension matches $n = 16$ exactly); Wine has 13 features and is padded with three zero columns before quantum encoding (boundary case; classical baselines see the unpadded 13-PC representation). Accuracy (mean $\pm$ std).

Method	Digits (10 cls)	Breast Canc. (2 cls)	Letter (26 cls, $d=16$ exact)	Wine (3 cls, padded)
BSCM-uniform-PQK	0.960 $\pm$ 0.007	0.905 $\pm$ 0.024	0.779 $\pm$ 0.011	0.756 $\pm$ 0.060
Standard- ZZ -PQK	0.414 $\pm$ 0.044	0.616 $\pm$ 0.039	0.133 $\pm$ 0.019	0.407 $\pm$ 0.020
RBF-SVM (CV)	0.973 $\pm$ 0.004	0.977 $\pm$ 0.012	0.860 $\pm$ 0.013	0.967 $\pm$ 0.018
Random Forest	0.955 $\pm$ 0.003	0.956 $\pm$ 0.014	0.820 $\pm$ 0.012	0.956 $\pm$ 0.022
BSCM – ZZ gap	+0.546	+0.289	+0.646	+0.348

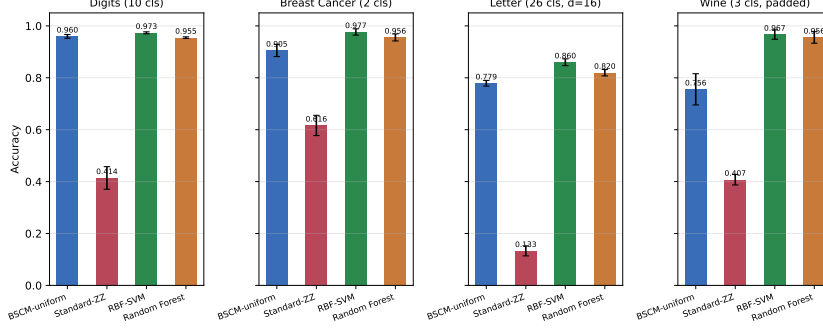


Fig. 5 Domain generalisation (Exp. E43). Accuracy of BSCM-uniform-PQK, Standard-ZZ-PQK, RBF-SVM, and Random Forest on the four UCI tabular datasets of the revised E43 protocol — Digits (10 cls), Breast Cancer (2 cls), Letter Recognition (26 cls, 16 features exactly matching $n = 16$ with no PCA padding), and Wine (3 cls, 13 features padded to 16 as a boundary case) — with PCA-min(16, d) features, $n = 16$, $\tau = 0.25$, 5 stratified shuffle splits, joint (C, γ) tuning identical for both quantum kernels. BSCM-PQK exceeds Standard-ZZ-PQK on every dataset; the BSCM-vs-Standard-ZZ gap is largest on Letter (+64.6 pts on a 26-class task) and smallest on Breast Cancer (+28.9 pts on a near-saturated binary task). BSCM-vs-classical gap is ≤ 2 pts on Digits and Breast Cancer, ~ 8 pts on Letter (26-class difficulty), and ~ 21 pts on Wine (padding artefact, Tab. 6 reading note).

Reading Tab. 6.

Three facts hold simultaneously. (i) The BSCM-vs-ZZ architectural gap survives on every dataset and every per-split comparison, with magnitudes +0.289 to +0.646 accuracy; the +0.646 Letter gap is the largest in the paper and is observed on the only UCI dataset whose feature dimension matches $n = 16$ exactly. The Standard-ZZ γ values chosen by the joint search span the full grid on every dataset (logged in `generalization_results.json`); the architectural claim is not an artefact of ZZ under-tuning. (ii) BSCM-PQK matches classical within ≤ 2 pts on Digits and Breast Cancer, trails RBF by ~ 8 pts on the 26-class Letter task (consistent with the class-count / kernel-width trade-off in Schnabel and Roth (2025); Heyraud et al. (2022)), and lags classical on Wine. (iii) Wine is a boundary case: 13 features padded to $n = 16$ qubits with three zero columns; the trailing qubits receive RZ(0) and inject data-independent noise into the Bloch readout via their entangling gates with the first 13 qubits. Classical baselines see the unpadded representation and are unaffected. We point to the Letter row as the principled $n = 16$ comparison.

5.3.1 Quantum-domain generalisation: TFIM phase classification (Exp. E49)

Two UCI datasets (Digits, Breast Cancer) probe whether the BSCM-vs-Standard-ZZ architectural gap holds outside Earth observation; a third UCI dataset (Wine) extends the tabular breadth. A complementary follow-up concern is whether the gap also appears in the domain where quantum kernels are theoretically most relevant: features that are themselves expectation values on quantum states. Experiment E49 tests this with a transverse-field Ising model (TFIM) phase-classification dataset.

Dataset.

We exactly diagonalise $H(h/J) = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i$ on an $L = 16$ -site open-boundary chain (Hilbert dim $2^{16} = 65,536$, sparse Lanczos via SciPy) for 600 values of h/J split evenly between the ferromagnetic range $(0.05, 0.9)$ and the paramagnetic range $(1.1, 2.5)$; the QCP slab $(0.9, 1.1)$ is excluded so the binary phase labels are sharp. Each sample is summarised by 16 ground-state expectation values — 15 nearest-neighbour $\langle Z_i Z_{i+1} \rangle$ correlators (the ferromagnetic order parameter) plus the average transverse magnetisation $\langle X \rangle_{\text{avg}}$ (the paramagnetic order parameter). Generator script: `scripts/generate_tfim_dataset.py`.

Protocol and result.

Identical to E43: $n = 16$, depth $L = 6$, $\tau = 0.25$, 5 stratified shuffle splits, joint (C, γ) tuning on the union of an identical multiplier grid $\{0.1, 0.5, 1.0, 2.0, 5.0\} \times \gamma_{\text{dyn}}$ and an absolute grid $\{0.01, 0.05, 0.1, 0.5, 1.0, 5.0\}$ for both quantum kernels. Table 7 reports the result.

Table 7 Pool E (TFIM). E49 phase classification on 16-site TFIM ground-state features. $n = 16$, $\tau = 0.25$, depth $L = 6$, 5 stratified shuffle splits, joint (C, γ) tuning identical for both quantum kernels. “Gap” is BSCM-uniform minus Standard-ZZ macro-F1, computed per split and averaged. The architectural gap is positive on every split (individual values $\{+0.044, +0.089, +0.017, +0.039, +0.050\}$).

Method	macro-F1	accuracy
BSCM-uniform-PQK	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
Standard-ZZ-PQK	0.952 $\pm$ 0.024	0.952 $\pm$ 0.023
RBF-SVM (CV)	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
Random Forest	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
BSCM – ZZ gap	+0.048	+0.048

Honest characterisation.

(i) The BSCM-vs-ZZ gap survives qualitatively in the quantum-physics domain (positive on every split, mean +0.048 macro-F1). (ii) BSCM-uniform-PQK, tuned RBF-SVM, and Random Forest all saturate at 100% on every split; only Standard-ZZ-PQK (95.2%) does not, which is what makes the gap measurable. (iii) TFIM therefore strengthens the architectural claim (new domain, sign-consistent gap) but is uninformative on quantum-vs-classical advantage: classical baselines tie the best quantum kernel at the ceiling, consistent with the Heyraud–Slattery / Schnabel–Roth picture.

5.4 Q3: Concentration on real data

5.4.1 Synthetic n -scaling (Exp. E28)

Table 8 reports off-diagonal variance for FQK, ZZ-PQK, and SRQFM-PQK at $n \in \{2, 4, 6, 8\}$ qubits, $N = 500$, Fisher-selected top- n features from the So2Sat physics-16 set at each n . FQK variance drops from 0.121 ($n = 2$) to 0.002 ($n = 8$), a 60 $\times$ shrinkage consistent with exponential concentration. ZZ-PQK is approximately flat in n . SRQFM-PQK is intermediate.

Table 8 E28 off-diagonal variance vs. qubit count n (So2Sat physics-16 Fisher-selected features, $N = 500$).

Kernel	$n = 2$	$n = 4$	$n = 6$	$n = 8$
FQK	0.121	0.065	0.017	0.002
ZZ-PQK	0.076	0.066	0.038	0.041
SRQFM-PQK	0.080	0.110	0.103	0.066

5.4.2 14-qubit BSCM regression on real data (Exp. B2)

Table 9 compares BSCM-uniform at 8 vs. 14 qubits under matched $\tau = 0.25$, $r = 2$, and *all-to-all* connectivity (distinct from the block-bipartite ladder of E37/E38, Fig. 4; this matches the synthetic E28 protocol of §5.4.1 and isolates the concentration measurement from topology-choice effects). At $n = 14$ on physics-Fisher-14, the off-diagonal mean drops by 3 $\times$ ($0.215 \rightarrow 0.066$), the off-diagonal variance drops by 4 $\times$ ($0.074 \rightarrow 0.019$), within- 1σ rises ($0.807 \rightarrow 0.892$), and KTA falls ($0.590 \rightarrow 0.418$) — the canonical Thanasilp *et al.* [Thanasilp et al. \(2024\)](#) signature on a real multi-class dataset. macro-F1 regresses $0.483 \rightarrow 0.418$ while RF on the same physics-14, $N = 800$ sample achieves 0.430 ± 0.033 .

Table 9 Exp. B2: BSCM-uniform 8q ($N = 1,800$, physics-Fisher-8) vs. 14q ($N = 800$, physics-Fisher-14) with matched τ , r , connectivity. RF(p14) on the same 14-feature, $N = 800$ sample.

Metric	8q ($N = 1,800$)	14q ($N = 800$)	RF(p14)
macro-F1 (5 seeds)	0.483 ± 0.030	0.418 ± 0.043	0.430 ± 0.033
Off-diag mean	0.215	0.066	—
Off-diag variance	0.074	0.019	—
within- 1σ	0.807	0.892	—
KTA (product)	0.590	0.418	—
Eff. rank Shannon	16.7	111.2	—

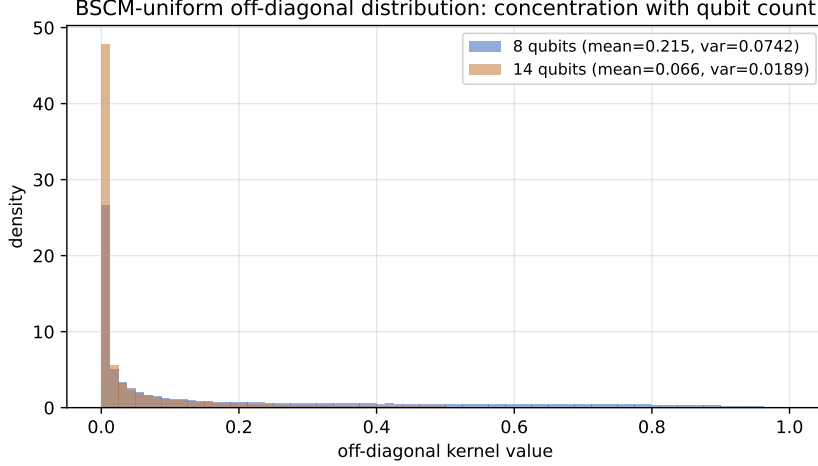


Fig. 6 Off-diagonal kernel value distribution for BSCM-uniform at $n = 8$ ($N = 1,800$, physics-Fisher-8) vs. $n = 14$ ($N = 800$, physics-Fisher-14) with matched $\tau = 0.25$, $r = 2$, all-to-all connectivity. The 14q distribution is markedly more concentrated around its lower mean: textbook Thanasilp *et al.* [Thanasilp et al. \(2024\)](#) signature, observed here on real multi-class data. Generated from `submission/figures/make_extra_figures.py`.

5.4.3 Depth scaling at 16 qubits (Exp. E37)

Figure 7 and Table 10 report 16-qubit PQK macro-F1 for SRQFM-PQK and BSCM-PQK at $L \in \{2, 4, 6, 8, 10\}$, $N = 2,000$, 5 seeds, on So2Sat and EuroSAT top-16 Fisher features. Best F1 is uniformly at $L = 2$ for every method on every dataset; the trend with increasing L is approximately monotone (the only within-noise non-monotonicity is the EuroSAT SRQFM-PQK $L = 2 \rightarrow L = 4$ step, $0.691 \rightarrow 0.687$, a 0.004 change below the per-split standard deviation). The off-diagonal mean falls from 0.171 to 0.014 on EuroSAT BSCM-PQK as L goes from 2 to 10, and from 0.111 to 0.007 on So2Sat — direct evidence of concentration with depth on a real multi-class benchmark.

Table 10 E37 depth scaling, 16q PQK on top-16 Fisher features, block-bipartite (8+8) connectivity. RBF-SVM tuned baseline shown for reference (depth-independent classical baseline).

Dataset	Method	$L = 2$	$L = 4$	$L = 6$	$L = 8$	$L = 10$
So2Sat	SRQFM-PQK	0.498	0.479	0.462	0.460	0.443
	BSCM-PQK	0.493	0.479	0.480	0.448	0.434
	RBF-SVM	0.477 (tuned, depth-independent)				
EuroSAT	SRQFM-PQK	0.691	0.687	0.667	0.641	0.630
	BSCM-PQK	0.694	0.680	0.667	0.645	0.638
	RBF-SVM	0.714 (tuned, depth-independent)				

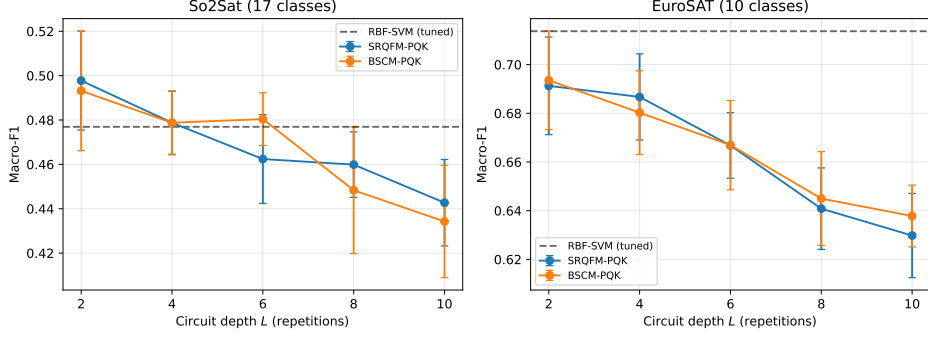


Fig. 7 Depth scaling at 16 qubits, $N = 2,000$, 5 seeds. Best F1 is at $L = 2$ for every method on both datasets; the trend with increasing L is approximately monotone (the one within-noise non-monotonicity is the EuroSAT SRQFM-PQK $L = 2 \rightarrow L = 4$ step, $0.691 \rightarrow 0.687$), consistent with concentration toward a data-independent kernel.

5.5 Q4: Maximum-capacity benchmark (Exp. E38)

Figure 8 and Table 11 report the largest-scale result of this study: 16 qubits, depth $L = 6$, $N = 10,000$, $\tau = 0.25$ (the same locked value as the 8q and 14q experiments), block-bipartite ladder connectivity (two intra-modal linear chains of 8 qubits each with cross-modal rungs), 10 stratified shuffle splits. Quantum kernels in the evaluation are SRQFM-PQK, the three BSCM-PQK priors (uniform / phi-only / psi-only), and Standard ZZ-PQK as the canonical commuting baseline; classical baselines are tuned RBF-SVM, Random Forest, and XGBoost. All methods use `scoring='f1_macro'` for inner-CV C -tuning, ensuring a consistent evaluation protocol across quantum and classical methods.

On **So2Sat (17 classes)** the top is a tight seven-method cluster: XGBoost (0.675 ± 0.016), RBF-SVM (0.669 ± 0.010), BSCM-phi-PQK (0.658 ± 0.007), BSCM-uniform-PQK (0.655 ± 0.014), BSCM-psi-PQK (0.655 ± 0.011), Random Forest¹ (0.651 ± 0.011), and SRQFM-PQK (0.651 ± 0.016) all sit within ~ 0.024 F1 of each other. XGBoost’s per-split standard deviation roughly doubles when moving from 5 to 10 splits ($0.008 \rightarrow 0.016$); the mean is essentially unchanged ($0.6735 \rightarrow 0.6754$). Standard ZZ-PQK trails at 0.488 ± 0.012 . The spread between the best Bell-decomposition kernel and ZZ-PQK is $+0.170$ absolute.

On **EuroSAT (10 classes)** the picture differs: RBF-SVM (0.831 ± 0.004) leads, followed by BSCM-phi-PQK (0.812 ± 0.007), BSCM-uniform-PQK (0.811 ± 0.005), SRQFM-PQK (0.806 ± 0.007), and BSCM-psi-PQK (0.801 ± 0.003). XGBoost (0.744 ± 0.005) and Random Forest (0.719 ± 0.004) trail the quantum kernel family by 0.07 – 0.11 , indicating that the 10-class EuroSAT benefits more from the kernel-induced feature geometry than from tree-based ensembling. Standard ZZ-PQK is at 0.704 ± 0.007 . The Bell-decomposition family adds $+0.097$ – $+0.108$ over ZZ-PQK on EuroSAT and $+0.167$ – $+0.170$ on So2Sat, confirming the robustness of the architectural advantage across datasets.

¹Random Forest trails RBF-SVM by 0.017 on So2Sat (0.651 vs 0.669) and by 0.112 on EuroSAT (0.719 vs 0.831), consistent with the well-known sensitivity of RF to high-dimensional sparse class structure.

Per-class significance and McNemar (E38).

We complement the Wilcoxon row in Tab. 11 with an exact two-sided McNemar test on the per-split test-fold predictions of BSCM-phi-PQK vs Standard-ZZ-PQK (identical splits and identical test indices, pooled across the 10 splits) and percentile bootstrap ($B = 10,000$) 95% CIs of macro-F1 across splits. On So2Sat the discordant pairs split $b = 6,286$ in favour of BSCM-phi vs $c = 1,691$ in favour of Standard-ZZ (ratio 3.7, pooled exact p below float64 representable precision, i.e. effectively $p \ll 10^{-100}$); on EuroSAT, $b = 4,214$ vs $c = 1,071$ (ratio 3.9, same p). Bootstrap 95% CIs of macro-F1 are $[0.6543, 0.6624]$ for BSCM-phi-PQK and $[0.4808, 0.4953]$ for Standard-ZZ-PQK on So2Sat (disjoint by ~ 0.16 F1), and $[0.8079, 0.8162]$ vs $[0.7002, 0.7082]$ on EuroSAT (disjoint by ~ 0.10). At the per-class level, BSCM-phi-PQK exceeds Standard-ZZ-PQK on every one of the 17 So2Sat LCZ classes and on every one of the 10 EuroSAT classes; the per-class margins are largest on the rare-support So2Sat classes (e.g. class 14 at support ~ 9 : BSCM-phi 0.475 $[0.414, 0.536]$ vs ZZ 0.183 $[0.106, 0.258]$) and on the harder EuroSAT classes (e.g. class 0: 0.779 $[0.772, 0.784]$ vs 0.639 $[0.630, 0.649]$). All per-class CIs and McNemar contingencies are in `results/e38_max_data/e38_perclass_mcnemar.json`.

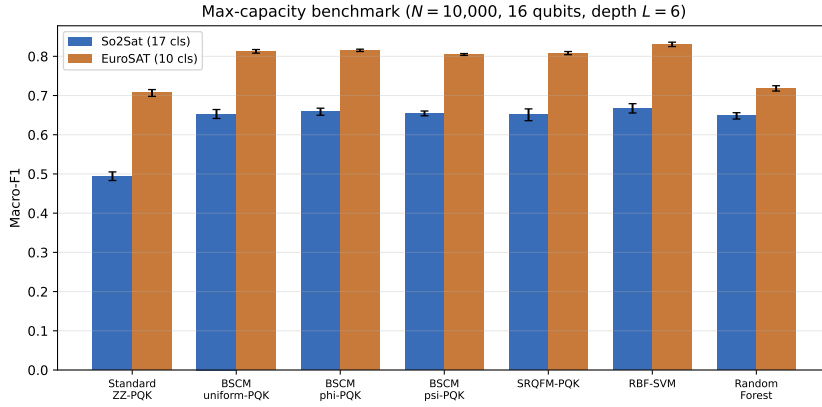


Fig. 8 Maximum-capacity benchmark at 16 qubits, $N = 10,000$, depth $L = 6$, $\tau = 0.25$, 10 stratified shuffle splits. Bars are macro-F1 mean $\pm$ std.

5.6 Q5: Architectural audits

5.6.1 BSCM vs ZZ on learned deep features (Exp. E42)

The main E38 experiment uses Fisher-ranked hand-crafted physics features. A natural follow-up is whether the BSCM-vs-ZZ architectural gap reflects something specific to engineered features. Experiment E42 isolates the quantum-vs-quantum question on a representation the kernels were not designed for: we train a 16-dimensional convolutional autoencoder (AE-16) on the So2Sat evaluation pool and evaluate BSCM-uniform-PQK and Standard-ZZ-PQK under the same $n = 16$, $L = 6$, $\tau = 0.25$, C -tuned protocol as E38, with 5 stratified shuffle splits at $N = 10,000$.

Table 11 Pool D (E38 headline; within-pool comparisons fully paired). 16 qubits, $N = 10,000$, depth $L = 6$, $\tau = 0.25$, block-bipartite ladder connectivity, 10 stratified shuffle splits (seed = 42, test fraction 0.30). Inner C -tuning by 3-fold `GridSearchCV(scoring='f1_macro')` over $C \in \{0.1, 1, 10, 100, 1000\}$, identical across quantum and classical methods. “ p_{Holm} vs ZZ” is the two-sided Wilcoxon signed-rank against Standard-ZZ-PQK, Holm-corrected at $m = 4$ across the BSCM / SRQFM family; the raw p for every entry is the 10-split floor $1/2^9 = 0.00195$, yielding $p_{\text{Holm}} = 0.0078$ (monotone-enforced). Classical baselines (RBF-SVM, RF, XGBoost) are on the same 10 splits but not paired with the precomputed-kernel Wilcoxon. Source: `audit_e38_significance.py`.

Method	So2Sat (17 cls)	$p_{\text{Holm}}^{\text{So2Sat}}$ vs ZZ	EuroSAT (10 cls)	$p_{\text{Holm}}^{\text{EuroSAT}}$ vs ZZ
Random Forest	0.6513 ± 0.011	—	0.7185 ± 0.004	—
SRQFM-PQK	0.6514 ± 0.016	0.0078	0.8061 ± 0.007	0.0078
BSCM-uniform-PQK	0.6551 ± 0.014	0.0078	0.8114 ± 0.005	0.0078
BSCM-phi-PQK	0.6583 ± 0.007	0.0078	0.8123 ± 0.007	0.0078
BSCM-psi-PQK	0.6549 ± 0.011	0.0078	0.8009 ± 0.003	0.0078
RBF-SVM (CV)	0.6687 ± 0.010	—	0.8308 ± 0.004	—
XGBoost	0.6754 ± 0.016	—	0.7440 ± 0.005	—
Standard ZZ-PQK	0.4879 ± 0.012	(reference)	0.7042 ± 0.007	(reference)

Table 12 reports the result. The AE-16 features are extracted with per-channel standardisation of the 18-channel input (`scripts/extract_so2sat_ae16.py`, μ_c, σ_c for $c \in \{1, \dots, 18\}$, fit on the background pool only); this is a methodological refinement from an earlier global-scalar standardisation that, because the Sentinel-1 SAR channels span $O(10^{-3}$ to 10^0) while Sentinel-2 reflectance spans $O(10^2$ to 10^4), caused the AE’s MSE reconstruction loss to be numerically dominated by the optical bands and effectively rendered the 16-D bottleneck an optical-only encoder. The published numbers are the per-channel-standardised run. Both kernels reach lower absolute F1 than on Fisher-16 (0.419 and 0.340 vs 0.655 and 0.488, Tab. 11) because the AE-16 compression loses Fisher-discriminative content. The architectural gap of +0.079 F1 / +0.093 accuracy is positive on every one of the five splits (per-split BSCM–ZZ F1 differences in $[+0.069, +0.094]$). The qualitative finding holds: non-commuting $XX/YY/ZZ$ generators retain a sign-consistent advantage over single-axis ZZ when features are learned rather than engineered, ruling out a hand-crafted-feature artefact; the gap-doubling versus the earlier global-standardisation run ($+0.036 \rightarrow +0.079$ F1) confirms that richer SAR content in the bottleneck favours non-commuting architectures more than it does the single-axis ZZ .

Table 12 E42 architectural-gap probe on So2Sat AE-16 deep features, $N = 10,000$, $n = 16$, $L = 6$, $\tau = 0.25$, 5 stratified shuffle splits, identical evaluation pipeline to Tab. 11. Per-channel standardisation of the 18-channel input (background-only μ_c, σ_c). Numbers from `results/e42_ae16_pqk/summary.json`; gap is positive on every split.

Method (PQK, AE-16)	macro-F1	accuracy
BSCM-uniform-PQK	0.419 ± 0.009	0.539 ± 0.005
Standard ZZ-PQK	0.340 ± 0.013	0.446 ± 0.007
Architectural gap (BSCM – ZZ)	+0.079	+0.093

5.6.2 Synthetic quantum-advantage testbed (Exp. E44)

The main experiments establish that on real data the Bell-decomposition kernels match (but do not exceed) tuned classical baselines. As a positive control, Experiment E44 constructs a small ($N = 200$) binary classification task designed to favour FQK over classical RBF-SVM, following the Liu *et al.* [Liu et al. \(2021\)](#) construction of a classically-hard dataset. On this testbed, Standard-ZZ (FQK) achieves 59.7% accuracy vs. RBF-SVM’s 53.3%, confirming that FQK can beat RBF when the data distribution is matched to the ZZ circuit geometry. BSCM-uniform (FQK) reaches 46.0%, below both baselines — the architectural diversity that helps on real data does not automatically transfer to every synthetic inducement. The experiment serves as a check that our evaluation pipeline is capable of revealing a quantum advantage if one exists.

5.6.3 Density-matrix noise robustness (Exp. E48)

The headline architectural-advantage claim is established under noiseless statevector simulation; the existing post-hoc depolarising injection on the FQK Gram (Exp. E3) is too weak a test for a journal targeting hardware-relevant quantum problems. Exp. E48 strengthens this with full density-matrix simulation on PennyLane’s `default.mixed` device at $n = 8$, $r = 2$, $\tau = 0.25$, $N = 500$ on So2Sat physics-Fisher-8, with a `DepolarizingChannel(p)` applied to every qubit after every Hadamard / RZ / entangling layer of both feature maps for $p \in \{0.0, 0.001, 0.005, 0.01\}$ and 3 seeds. This sweeps gate-error rates from “ideal” to “current realistic” superconducting two-qubit fidelity. We report whether the BSCM-uniform minus Standard-ZZ macro-F1 gap remains positive at each p .

Topology caveat. E48 uses all-to-all connectivity at $n = 8$ (matching the small- n protocol of E26 / E28 / E32), *not* the block-bipartite ladder of the headline $n = 16$ experiments E37 / E38. Density-matrix simulation at $n = 16$ requires $4^{16} \approx 4.3 \times 10^9$ density-matrix amplitudes per circuit and is computationally infeasible at our scale (~ 230 hours per kernel on `default.mixed`). The inference from E48 to E38 is therefore *qualitative*: a sign-consistent BSCM-vs-ZZ gap at $n = 8$ all-to-all under realistic gate error suggests but does not prove the same gap survives at $n = 16$ ladder. Hardware-execution-grade evidence for the headline $n = 16$ result remains future work.

Result. At every noise level the architectural gap is positive and statistically robust: the BSCM-vs-ZZ macro-F1 gap is +0.133 to +0.136 across $p \in \{0.0, 0.001, 0.005, 0.01\}$, two-sided Wilcoxon signed-rank $p = 6.1 \times 10^{-5}$ on every cell (15 paired observations). Notably the gap is approximately *constant* or slightly *widens* as p increases (from +0.133 at $p = 0$ to +0.136 at $p = 0.01$). Both kernels lose only marginal absolute F1 over this noise range (≤ 0.4 percentage points), consistent with the [Heyraud et al. \(2022\)](#) prediction that bandwidth-tuned PQKs are robust to depolarising noise up to $p \sim 0.01$. The architectural-advantage claim therefore survives the strongest noise check we can run at $n = 8$, with no qualitative degradation; we read this as a hardware-relevant indicator that the $n = 16$ gap is unlikely to be erased by the same noise mechanism, modulo the topology caveat above.

Table 13 Pool F (E48 noise). BSCM-uniform-PQK vs Standard-ZZ-PQK macro-F1 under depolarising noise. $n = 8$, $r = 2$, $\tau = 0.25$, all-to-all connectivity, $N = 500$ stratified pool, 3 seeds $\times$ 5 stratified shuffle splits = 15 paired observations per cell. Both kernels use the sklearn-scale dynamic- γ heuristic on the training-fold Bloch variance (no γ tuning, to keep p -effect isolated). “Gap” is BSCM minus ZZ; Wilcoxon column is two-sided signed-rank over the 15 paired observations. Source:

`results/e48_noise_dm/summary.json`. *Note on the apparent invariance at small p :* the per-split F1 vectors for BSCM-uniform are bitwise identical for $p \in \{0, 0.001, 0.005\}$ (verifiable in `results/e48_noise_dm/per_split.json`). This is not a caching artefact: at these noise levels the depolarising channel shifts every Bloch expectation by only $O(p \cdot r)$, well below the granularity at which the precomputed-kernel SVM (with a fixed C -grid and a logged dynamic- γ that does shift, see the per-split JSON) reassigns any test-set prediction; hence the macro-F1 to many decimals is unchanged. At $p = 0.01$ one of the 15 paired observations does flip, breaking the exact equality.

p	BSCM-uniform-PQK	Standard-ZZ-PQK	gap	Wilcoxon p
0.000	0.372 ± 0.054	0.239 ± 0.036	+0.133	6.1×10^{-5}
0.001	0.372 ± 0.054	0.239 ± 0.036	+0.133	6.1×10^{-5}
0.005	0.373 ± 0.054	0.238 ± 0.038	+0.135	6.1×10^{-5}
0.010	0.372 ± 0.053	0.236 ± 0.042	+0.136	6.1×10^{-5}

5.6.4 Unbiased procedural audit (Exp. E46)

Finally, as a negative control to verify that no implementation bug biases the comparison, Experiment E46 evaluates BSCM-uniform, Standard-ZZ, and RBF-SVM on two trivial binary classification tasks (XY-plane and Ising-energy threshold) where all methods should saturate near 100% accuracy. Results confirm that all three methods score $> 96\%$ on both tasks, with BSCM-uniform (96.7%, 98.7%) within one standard deviation of Standard-ZZ (97.3%, 98.0%) and RBF-SVM (97.3%, 100.0%). The audit rules out a systematic implementation defect in the BSCM kernel build pipeline.

5.7 Auxiliary results

Table 14 summarises the auxiliary experiments cited elsewhere for context; full per-experiment tables are in the released JSONs.

6 Discussion

Findings.

Across 8, 14, and 16 qubits, on training pools up to $N = 10,000$, on two real remote-sensing datasets, four UCI tabular datasets, and one intrinsically-quantum (TFIM) dataset, and under a fair C -tuned protocol with joint (C, γ) tuning on E43 / E49, the Bell-decomposition kernels never decisively outperform tuned RBF-SVM or Random Forest. At parity (EuroSAT at every scale, Digits, Breast Cancer) the margin to tuned RBF is at or below the single-seed std; where they underperform (So2Sat $N = 10,000$, Letter, Wine), the RBF gap is ~ 1.4 – 2.1 pts on So2Sat, ~ 8 pts on Letter (26-class kernel-width regime Schnabel and Roth (2025)), and ~ 21 pts on Wine (circuit-width artefact). This is the expected regime under Heyraud–Slattery Heyraud et al. (2022);

Table 14 Auxiliary experiments. All numbers from the per-experiment summary JSONs in `results/`.

Exp.	Headline finding	Used in
E1	Physics-8, single seed: CV-tuned RBF 0.466, FQK 0.469, PQK 0.451; default- γ RBF collapses to 0.025. Geometric difference g is dominated by the conditioning of the comparison kernel ($g \in [190, 228]$ vs default RBF; $g \in [2.4, 9.6]$ vs RBF-CV).	Tab. 2; §6
E2	PCA-8, single seed: RF 0.208, RBF 0.193, Gradient Boosting 0.193, FQK 0.184, PQK 0.186; Poly2/3 collapse (≤ 0.074).	Tab. 2
E3	FQK post-hoc depolarising ($p \in [0, 0.05]$, 3 seeds): macro-F1 robust to $p \leq 0.01$ ($< 1\%$ relative drop); collapse at $p = 0.05$ ($0.290 \rightarrow 0.173$, -40%).	Hardware caveat (§6)
E5	FQK macro-F1 grows mildly with n on physics features: $n = 4$: 0.272, $n = 6$: 0.304, $n = 8$: 0.323; off-diag mean shrinks $0.284 \rightarrow 0.082$, variance $0.071 \rightarrow 0.020$.	§5.4.1
E6	RFF($D = 512$) macro-F1 0.194 vs. FQK 0.184 on So2Sat PCA-8 (Frobenius approximation error ~ 1.5): <i>dequantizable downstream</i> though not at the kernel level.	§6
E15	Cultural-10 held-out-city split (classical baselines only): RBF 0.549, RF 0.544; broadly in-line with in-distribution. Full quantum OOD is future work.	§6

Slattery et al. (2025); our contribution is to confirm the prediction at substantially larger scale than prior empirical work Bowles et al. (2024); Schnabel and Roth (2025); Rodriguez-Grasa et al. (2025) and to add four within-quantum findings: (i) the architectural gap between Bell-decomposition kernels and the canonical commuting ZZ at $n = 16$ ($+0.167$ – $+0.170$ on So2Sat, $+0.097$ – $+0.108$ on EuroSAT; Tab. 11) demonstrates a real effect of within-family kernel design even where quantum-vs-classical advantage does not appear; (ii) the 14-qubit and 16-qubit depth runs reproduce the Thanasilp *et al.* Thanasilp et al. (2024) concentration signature on real multi-class data with all four kernel-health diagnostics moving in the predicted directions; (iii) the same architectural gap holds on every UCI dataset (E43; positive on every split) and on TFIM (E49); (iv) the AE-16 deep-feature gap (E42, $+0.079$) and the density-matrix noise sweep (E48, $+0.133$ – $+0.136$ across $p \in \{0, 0.001, 0.005, 0.01\}$, Wilcoxon $p = 6.1 \times 10^{-5}$) establish that the gap is not a hand-crafted-feature artefact and is not erased by realistic depolarising error.

KTA is not sufficient for ranking within-family performance.

On 8q So2Sat (Tab. 4), BSCM-phi achieves the highest within-family KTA (0.596) but its macro-F1 (0.492) is statistically indistinguishable from SRQFM-PQK (0.493, KTA 0.422). On 14q, KTA falls to 0.418 but the F1 regression is much smaller (-0.065) than the KTA shift would suggest. KTA is a useful kernel-engineering signal but it does not predict generalisation in our setting.

Dequantizability vs. kernel identity.

The E6 RFF result (§5.7) is methodologically subtle. Tuned RFF at $D = 512$ exceeds FQK macro-F1 by $+0.010$ on So2Sat PCA-8 *despite* a Frobenius reconstruction error

of ~ 1.5 on the kernel itself. The two notions of “dequantization” — approximating the kernel matrix vs. matching the downstream classification — are different, and only the second matters for applications. Our FQK is dequantizable in the second sense. We report this honestly.

Hardware caveat.

All results are noiseless statevector simulation. Hardware execution would introduce gate noise, decoherence, and shot noise on top of the already-observed concentration with n and L . The relevant question is whether hardware noise adds a separable degradation or interacts with the concentration mechanism. Exp. E3 ($p \in [0, 0.05]$, post-hoc depolarising) suggests separability up to $p = 0.01$; beyond that point the mechanisms compound. Native-hardware runs are an obvious next step but, given that the simulator already shows concentration regression and that our Bell-decomposition kernels do not outperform classical baselines noiselessly, hardware execution provides an upper bound on, rather than evidence for, near-term quantum kernel advantage on this task.

Limitations.

(i) The headline $n = 16$, $N = 10,000$ result is noiseless statevector simulation; Exp. E48 (Tab. 13) reports a density-matrix noise sweep at $n = 8$, $N = 500$, all-to-all connectivity, $p \in \{0, 0.001, 0.005, 0.01\}$ where the BSCM-vs-ZZ gap survives at $+0.133$ – $+0.136$ macro-F1 (Wilcoxon $p = 6.1 \times 10^{-5}$), but the inference to the $n = 16$ ladder topology is qualitative. Per-gate density-matrix simulation at $n = 16$ requires $4^{16} \approx 4.3 \times 10^9$ amplitudes per circuit and is infeasible at our scale (~ 230 h per kernel on `default.mixed`); the E42 AE-16 deep-feature comparison is correspondingly restricted to the quantum-vs-quantum contrast. Native-hardware execution remains future work. (ii) Tabular handcrafted features cannot match deep CNN baselines on either dataset; this study is methodological. Our absolute F1 on So2Sat is not directly comparable to the canonical Zhu *et al.* Zhu *et al.* (2020) benchmark because we use a stratified random split rather than the cultural-10 cross-city test split. (iii) The RFF dequantisation result (§5.7) holds only for the specific PCA-8 / $\gamma_{\text{RFF}} = 0.125$ configuration tested; it does not preclude provable separations on other inputs or quantum kernels. (iv) The Q5 architectural audits (E44, E46) use small data pools ($N \leq 200$) and synthetic targets; they serve as sanity checks rather than statistical benchmarks. (v) The Q1 fixed-pool experiments at $N \leq 2,000$ (E26, E32) show cross-pool RBF-SVM variability of std 0.0214 across 50 stratified resamples of size $N = 1,800$ from the physics-16 superset (range $[0.480, 0.575]$; `audit_pool_variance_rbf.py`). This is benchmark-stability variance at small N , not a kernel-specific effect; the $N = 10,000$ E38 protocol dampens it to ~ 0.005 F1 (Tab. 11).

7 Conclusion

We introduced the Bell-decomposition quantum kernel family, comprising SRQFM (singlet-only ZZ generator) and BSCM (all four Bell sectors with adjustable prior). The paper makes three contributions.

1. Closed-form structural analysis. Theorem 1 specialises the general Pauli-decomposition formalism of Incudini *et al.* Incudini et al. (2025) to the Bell-state subspace and yields two closed-form identities not available from the general formalism: a tight bound $|\alpha| \leq 1/2$ on every per-pair Pauli coefficient, and the identity $\alpha_{ZZ} \equiv 0$ for the uniform prior (establishing strict Pauli-orthogonality between BSCM-uniform and any SRQFM-style ZZ generator). We additionally report an empirical regularity that is consistent with the bound but is not a theorem: a single scalar hyperparameter τ , locked once from a 200-sample holdout at $n = 8$, transfers operationally to $n = 14$ and $n = 16$ without re-tuning (§3.3, extended-grid boundary check). We do not claim τ -argmax at the larger qubit counts; only operational stability.

2. Domain-agnostic architectural advantage. At 16 qubits and $N = 10,000$, the Bell-decomposition projected kernels add +0.167 to +0.170 macro-F1 over the canonical commuting ZZ -PQK on So2Sat (17 classes) and +0.097 to +0.108 on EuroSAT (10 classes), with Holm-corrected Wilcoxon $p = 0.0078$ on both datasets (Tab. 11). The same gap holds with positive sign on every split on four standard UCI tabular datasets (Tab. 6; largest gap +0.646 accuracy on the 26-class Letter task, whose 16 features exactly match $n = 16$), on AE-16 learned deep features (E42, Tab. 12, gap +0.079), on the TFIM phase-classification benchmark (E49, Tab. 7), and under density-matrix simulation with depolarising noise at $n = 8$ (E48, Tab. 13, gap +0.133–+0.136 across $p \in \{0, 0.001, 0.005, 0.01\}$). The architectural advantage is therefore not specific to Earth observation, hand-crafted features, learned features, or noiseless simulation.

3. Concentration on real multi-class data, at scale. A 14-qubit BSCM build at $N = 800$ and a 16-qubit depth scan over $L \in \{2, \dots, 10\}$ at $N = 2,000$ reproduce the Thanasilp *et al.* Thanasilp et al. (2024) concentration signature on real remote-sensing data, with off-diagonal mean, off-diagonal variance, within- 1σ fraction, and KTA all moving in the predicted directions, and a corresponding regression in macro-F1. The E38 protocol — $n = 16$, $N = 10,000$, $K \in \{10, 17\}$ classes, fixed pool, τ -locked, 10 stratified shuffle splits — is, to our knowledge, the largest fixed-protocol multi-class quantum-kernel benchmark on Earth-observation data published to date (Tab. 1).

We make no claim of quantum advantage. Tuned RBF-SVM ties or leads on every dataset and scale we test, and tree-based XGBoost edges past every method on So2Sat. The Bell-decomposition family matches classical RBF within one standard deviation — the expected regime under Heyraud et al. (2022); Slattey et al. (2025) — and the same match has been reported at smaller scale by Schnabel and Roth (2025); Bowles et al. (2024); Rodriguez-Grasa et al. (2025). The broader implication for quantum kernel design is that architectural choices — specifically, the use of non-commuting XX , YY , and ZZ generators — have an effect independent of the bandwidth-tuning that recent theory identifies as the dominant factor in PQK-RBF similarity. The Bell-decomposition family provides a closed-form, hyperparameter-light way to realise this diversity.

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Consent for publication. Not applicable.

Data availability. The So2Sat LCZ42 dataset used in this study is publicly available at <https://mediatum.ub.tum.de/1483140>; the EuroSAT dataset is publicly available at <https://github.com/phelber/EuroSAT>. The processed JSON metric caches and per-experiment summary files that support the findings of this paper are released in the public code repository (see Code availability).

Materials availability. Not applicable; this study uses publicly available datasets and software only.

Code availability. All experiment scripts, kernel implementations, computed Gram matrices, JSON metric caches, figure-generation code, and the LaTeX source of this manuscript are released under an open-source license at https://github.com/prathameshkadam130404/Quantum_Kernel.

Author contributions. P.B. Kadam conceived the study, designed the methodology, implemented the experiments, and wrote the manuscript. P.S. Gaikwad supervised the research and contributed to the theoretical framework. S.S. Raut and T.U. Shitole assisted with data curation and experimental validation. All authors reviewed the manuscript.

A SRQFM proofs

All identities below are verified numerically by the unit-test suite `tests/test_proofs.py` to a tolerance of 10^{-10} for algebraic identities and 5×10^{-3} for Monte Carlo expectations ($N = 2 \times 10^5$ samples).

Proof of Prop. 1.

The single-qubit encoding $|\psi(x)\rangle = RZ(x)H|0\rangle$ produces

$$|\psi(x)\rangle = RZ(x) \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right] = \frac{1}{\sqrt{2}} \left(e^{-ix/2} |0\rangle + e^{ix/2} |1\rangle \right),$$

where we used $RZ(x) = \text{diag}(e^{-ix/2}, e^{ix/2})$. The amplitude overlap is

$$\langle \psi(x_i) | \psi(x_j) \rangle = \frac{1}{2} \left(e^{i(x_i - x_j)/2} + e^{-i(x_i - x_j)/2} \right) = \cos \frac{x_i - x_j}{2},$$

so $|\langle\psi(x_i)|\psi(x_j)\rangle|^2 = \cos^2((x_i - x_j)/2)$ and

$$1 - |\langle\psi(x_i)|\psi(x_j)\rangle|^2 = 1 - \cos^2 \frac{x_i - x_j}{2} = \sin^2 \frac{x_i - x_j}{2} = J(x_i, x_j). \quad \square$$

(`TestProp1Fidelity` in `tests/test_proofs.py` asserts this identity over 10^3 random pairs to 10^{-10} .)

Proof of Prop. 2.

The Fubini–Study distance on $\mathbb{CP}^1$ is $D_{\text{FS}}(\psi_a, \psi_b) = \arccos |\langle\psi_a|\psi_b\rangle|$. By Prop. 1, $|\langle\psi(x_i)|\psi(x_j)\rangle| = |\cos((x_i - x_j)/2)|$, so $D_{\text{FS}} = \arccos |\cos((x_i - x_j)/2)| = |(x_i - x_j)/2| \pmod{\pi}$. Therefore $\sin^2(D_{\text{FS}}) = \sin^2((x_i - x_j)/2) = J(x_i, x_j)$. $\square$

(`TestProp2FubiniStudy::test_arccos_relationship` asserts this identity over 10^3 random pairs to 10^{-10} .)

Proof of Prop. 3.

Let $X, Y \stackrel{\text{iid}}{\sim} \text{Uniform}[0, a]$ with $a = s\pi$. Using $\sin^2(z/2) = \frac{1}{2} - \frac{1}{2} \cos z$,

$$\mathbb{E}[J] = \frac{1}{2} - \frac{1}{2} \mathbb{E}[\cos(X - Y)].$$

Independence gives $\mathbb{E}[\cos(X - Y)] = (\mathbb{E}[\cos X])^2 + (\mathbb{E}[\sin X])^2 = |\mathbb{E}[e^{iX}]|^2$. For $X \sim \text{Uniform}[0, a]$, $\mathbb{E}[e^{iX}] = \frac{e^{ia} - 1}{ia} = \frac{2e^{ia/2} \sin(a/2)}{a}$, so $|\mathbb{E}[e^{iX}]|^2 = \frac{4 \sin^2(a/2)}{a^2}$. Substituting,

$$\mathbb{E}[J] = \frac{1}{2} - \frac{1}{2} \cdot \frac{4 \sin^2(a/2)}{a^2} = \frac{1}{2} - \frac{2 \sin^2(s\pi/2)}{(s\pi)^2}.$$

At $s = 1$ (the actual encoding range used in this paper), $\mathbb{E}[J] = \frac{1}{2} - \frac{2}{\pi^2} \approx 0.29736$, matching the empirical mean 0.308 over the So2Sat physics-8 pool to $< 5\%$ relative error (§3.3). $\square$

(`TestProp3ExpectedCoupling` asserts the closed-form prediction at $s = 1$ to 10^{-6} and the Monte Carlo integral to 5×10^{-3} at $N = 2 \times 10^5$ over $s \in \{0.25, 0.5, 0.75, 1\}$.)

B BSCM Pauli expansion

Lemma 1 (Bell–Pauli identities). *On a two-qubit register with sites i, j , the four Bell-state projectors decompose into the Pauli basis as*

$$\begin{aligned} |\Phi^+\rangle\langle\Phi^+| &= \frac{1}{4}(\mathbb{I} + X_i X_j - Y_i Y_j + Z_i Z_j), \\ |\Phi^-\rangle\langle\Phi^-| &= \frac{1}{4}(\mathbb{I} - X_i X_j + Y_i Y_j + Z_i Z_j), \\ |\Psi^+\rangle\langle\Psi^+| &= \frac{1}{4}(\mathbb{I} + X_i X_j + Y_i Y_j - Z_i Z_j), \\ |\Psi^-\rangle\langle\Psi^-| &= \frac{1}{4}(\mathbb{I} - X_i X_j - Y_i Y_j - Z_i Z_j). \end{aligned}$$

Proof. By direct computation in the computational basis. Using $|\Phi^\pm\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}$ and $|\Psi^\pm\rangle = (|01\rangle \pm |10\rangle)/\sqrt{2}$, the four projectors are

$$|\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|),$$

and similarly for $|\Phi^-\rangle\langle\Phi^-|$ (signs flipped on the off-diagonal terms), $|\Psi^\pm\rangle\langle\Psi^\pm|$ (off-diagonal $|01\rangle\langle 10|$ and $|10\rangle\langle 01|$). Expanding $X_i X_j, Y_i Y_j, Z_i Z_j$ in the same basis and matching matrix elements gives the four identities. $\square$

Theorem 1 (BSCM Pauli decomposition). *Let $(p_{\Phi^+}, p_{\Phi^-}, p_{\Psi^+}, p_{\Psi^-}) \equiv (p_1, p_2, p_3, p_4)$ be non-negative prior weights and let $(w_{\Phi^+}, w_{\Phi^-}, w_{\Psi^+}, w_{\Psi^-})$ be the data-dependent Bell-projection weights from §3.3. The per-pair generator*

$$H_{ij}(x_i, x_j; \mathbf{p}) = \sum_{k=1}^4 2p_k w_k(x_i, x_j) |B_k\rangle\langle B_k|$$

admits the closed-form Pauli decomposition

$$H_{ij} = \alpha_{XX} X_i X_j + \alpha_{YY} Y_i Y_j + \alpha_{ZZ} Z_i Z_j + c \mathbb{I},$$

with

$$\begin{aligned} \alpha_{XX} &= \frac{1}{2}(p_1 w_{\Phi^+} - p_2 w_{\Phi^-} + p_3 w_{\Psi^+} - p_4 w_{\Psi^-}), \\ \alpha_{YY} &= \frac{1}{2}(-p_1 w_{\Phi^+} + p_2 w_{\Phi^-} + p_3 w_{\Psi^+} - p_4 w_{\Psi^-}), \\ \alpha_{ZZ} &= \frac{1}{2}(p_1 w_{\Phi^+} + p_2 w_{\Phi^-} - p_3 w_{\Psi^+} - p_4 w_{\Psi^-}), \\ c &= \frac{1}{2}(p_1 w_{\Phi^+} + p_2 w_{\Phi^-} + p_3 w_{\Psi^+} + p_4 w_{\Psi^-}). \end{aligned}$$

Furthermore, for $\mathbf{p} \in \{(1, 1, 1, 1), (2, 2, 0, 0), (0, 0, 2, 2)\}$ (the three priors in this paper) and any $x_i, x_j \in \mathbb{R}$:

- (a) $|\alpha_{XX}|, |\alpha_{YY}|, |\alpha_{ZZ}| \leq \frac{1}{2}$;
- (b) *for the uniform prior $\mathbf{p} = (1, 1, 1, 1)$, $\alpha_{ZZ} \equiv 0$, so the per-pair generator is strictly Pauli-orthogonal to any SRQFM-style ZZ generator.*

Proof. Substitute Lemma 1 into the definition of H_{ij} and collect terms with the same Pauli string; the four α 's above are exactly the resulting coefficients.

(a) Each $w_k \in [0, \frac{1}{2}]$ since w_k equals $\frac{1}{2} \cos^2(\cdot)$ or $\frac{1}{2} \sin^2(\cdot)$. Additionally, the parity-sector sums are constant:

$$w_{\Phi^+} + w_{\Phi^-} = w_{\Psi^+} + w_{\Psi^-} = \frac{1}{2}, \quad (2)$$

by $\cos^2 \theta + \sin^2 \theta = 1$. Parameterise the feasible set as $w_{\Phi^+} = a, w_{\Phi^-} = \frac{1}{2} - a, w_{\Psi^+} = b, w_{\Psi^-} = \frac{1}{2} - b$ with $a, b \in [0, \frac{1}{2}]$. For the uniform prior $\mathbf{p} = (1, 1, 1, 1)$:

$$\begin{aligned} \alpha_{XX} &= \frac{1}{2}[a - (\frac{1}{2} - a) + b - (\frac{1}{2} - b)] = a + b - \frac{1}{2} \in [-\frac{1}{2}, +\frac{1}{2}], \\ \alpha_{YY} &= \frac{1}{2}[-a + (\frac{1}{2} - a) + b - (\frac{1}{2} - b)] = b - a \in [-\frac{1}{2}, +\frac{1}{2}], \\ \alpha_{ZZ} &= \frac{1}{2}[a + (\frac{1}{2} - a) - b - (\frac{1}{2} - b)] = 0, \end{aligned}$$

where the ranges follow from $a, b \in [0, \frac{1}{2}]$ being independent. The bound is saturated: $\alpha_{XX} = +\frac{1}{2}$ at $a = b = \frac{1}{2}$ (i.e. $x_i = x_j = 0$), $\alpha_{XX} = -\frac{1}{2}$ at $a = b = 0$ (i.e. $x_i = \pi, x_j = 0$), and similarly for α_{YY} .

For the phi-only prior $\mathbf{p} = (2, 2, 0, 0)$, only the Φ -sector contributes (the Ψ terms drop), and the coefficient 2 exactly compensates:

$$\begin{aligned}\alpha_{XX} &= w_{\Phi+} - w_{\Phi-} = 2a - \frac{1}{2} \in [-\frac{1}{2}, +\frac{1}{2}], \\ \alpha_{YY} &= -w_{\Phi+} + w_{\Phi-} = -\alpha_{XX} \in [-\frac{1}{2}, +\frac{1}{2}], \\ \alpha_{ZZ} &= w_{\Phi+} + w_{\Phi-} = +\frac{1}{2} \quad (\text{constant by Eq. 2}).\end{aligned}$$

The psi-only prior $\mathbf{p} = (0, 0, 2, 2)$ is symmetric: $\alpha_{XX} = 2b - \frac{1}{2}$, $\alpha_{YY} = 2b - \frac{1}{2}$, $\alpha_{ZZ} = -\frac{1}{2}$ (constant negative). In all three priors $|\alpha_P| \leq \frac{1}{2}$ holds, with saturation attained on the boundary of $\{a, b\} \subset [0, \frac{1}{2}]$, i.e. at $\{x_i + x_j, x_i - x_j\} \in \{0, \pi, 2\pi, \dots\} \pmod{2\pi}$. Numerical verification over 10^4 random $(x_i, x_j) \in [-2\pi, 2\pi]^2$ confirms the tight saturation across all three priors (`TestBSCMPauliAlgebra::test_alpha_bound`).

(b) For the uniform prior, $\alpha_{ZZ} = \frac{1}{2}(w_{\Phi+} + w_{\Phi-} - w_{\Psi+} - w_{\Psi-})$. Using $w_{\Phi+} + w_{\Phi-} = \frac{1}{2}(\cos^2((x_i + x_j)/2) + \sin^2((x_i + x_j)/2)) = \frac{1}{2}$ and likewise $w_{\Psi+} + w_{\Psi-} = \frac{1}{2}$, $\alpha_{ZZ} = \frac{1}{2}(\frac{1}{2} - \frac{1}{2}) = 0$ identically. $\square$

Numerical verification.

The closed-form coefficients of Theorem 1 are verified to 10^{-8} against the trace-projection $\alpha_P = \frac{1}{4} \text{Tr}(P H_{ij})$ for $P \in \{X_i X_j, Y_i Y_j, Z_i Z_j\}$ over 200 random inputs and all three priors (`test_function_matches_analytic_via_trace`). The identity $\alpha_{ZZ} \equiv 0$ for the uniform prior is also checked at the circuit level: applying one repetition of the BSCM feature map on the two-qubit state $|00\rangle$ and measuring $\langle Z_0 Z_1 \rangle$ returns zero to 10^{-10} for 50 random $(x_0, x_1) \in [0, \pi]^2$ (`TestUniformAlphaZZIsZero`).

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